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Assessing the Current People Counting Practice at the Cebu City Sports Center

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CHAPTER 1

INTRODUCTION

1.1 Introduction

People counting refers to the systematic measurement of determining the number of individuals occupying or moving through a space over time. At jogging facilities like the Cebu City Sports Complex (CCSC), people counting involves monitoring how many individuals enter, which lane they use, and their movement patterns within the oval. According to Pronello & Garzon Ruiz (2023), accurate monitoring is essential for maintaining operational efficiency, evaluating space utilization, and producing reliable data to guide evidence-based planning. Facilities traditionally relied on manual methods such as ticket stubs, tally sheets, and headcounts. However, these approaches are inefficient for real-time monitoring, prone to human error, and unable to capture lane-specific or peak-hour patterns, resulting in incomplete records (Fiandeiro et al., 2023).

The CCSC, widely known as Abellana, is one of Cebu City's most prominent public sports venues, serving both organized events and daily recreational use (Padronia & Lagunda, 2024). It is a jogging oval that consistently accommodates large volumes of users. However, current monitoring practices remain limited to approximate daily totals, leaving essential details about hourly peaks and lane distribution unrecorded (Seblos, 2024). The gap in monitoring underscores the need to closely examine how joggers use the facility, lane by lane.

Table 1. Cebu City Sports Complex Daily Visitor Data and Operational

Date	Daily Visitor Count (Joggers in Oval)	Operational Status / Remarks
JUNE 24, 2025	1,585 people	OPEN
JUNE 25, 2025	1,232 people	OPEN
JUNE 26, 2025	1,564 people	OPEN
JUNE 27, 2025	1,287 people	OPEN
JUNE 28, 2025	1,162 people	OPEN
JUNE 29, 2025	CLOSED	CFC EVENT
JUNE 30, 2025	490 people	OPEN

The research team coordinated with CCSC staff who manage jogger ticket sales records to acquire the data presented in Table 1. During the data request, the researchers observed that attendance records were maintained as manual ticket sales logs, stored in filing cabinets, and tallied using calculators. Staff members manually computed totals by going through multiple pages of daily ticket records. Instead of providing exact daily attendance figures, they initially offered an approximate monthly total of 30,000 joggers. When the researchers requested per-day data, the personnel

explained that retrieving those details would take considerable time because of the system's manual setup. After several follow-ups, the staff provided approximate daily attendance figures via text message, but these figures lacked supporting documentation or verifiable tallies. Consequently, the available dataset covers only one week, from June 24 to June 30, 2025, and consists of daily totals based on jogger pass ticket sales, without breakdowns by hour or specific time intervals. This reveals that while general attendance is recorded, the system currently fails to maintain systematic monitoring of jogger flow and time-based usage patterns.

The CCSC oval continues to rely on traditional methods, particularly ticketing, to estimate user numbers (Seblos, 2025). Staff issue passes to joggers, but certain groups, such as gym members with free access and students, are not regularly included in the count. The existing practices leave a significant portion of daily users unrecorded. The process of retrieving past data is also inefficient, as staff must sort through cabinets and manually add up numbers, delaying access to accurate information. Such practices make it difficult to analyze trends promptly and highlight the limitations of relying solely on paper-based tracking.

The objective of this research is to examine the current personnel counting system at the CCSC and evaluate its effectiveness in recording jogger attendance. Specifically, the study aims to document the methods currently used to track entries, identify their limitations, and determine how occupancy is monitored at different times of day, particularly during peak hours. Based on these findings, the study will propose

recommendations for more efficient people-counting approaches that provide timely, accurate data suited to the CCSC's operational context. The importance of this research lies in its capacity to enhance facility management by ensuring that attendance records are reliable and readily accessible.

1.2 Theoretical Background

Accurate and efficient people counting is essential for understanding space usage, ensuring facility compliance, and supporting data-driven decision-making in high-traffic public venues. As this study aims to assess the current people-counting practices used at the Cebu City Sports Complex (CCSC), it is essential to ground the research in theoretical concepts that explain the spatial and flow-based aspects of counting individuals in shared environments. The following theories provide the foundation for analyzing how people counting systems operate and how effectively they reflect actual occupancy.

Occupancy Density Theory (Alfalah, 2022) is a foundational concept in crowd dynamics and environmental design that examines the relationship between the number of individuals in a given area and the available space per person. Expressed in metrics such as people per square meter, this theory provides a quantitative basis for understanding how crowding affects comfort, movement, and overall usability of a space. It is frequently applied in architecture, urban planning, and crowd management to design and manage environments that accommodate people effectively and sustainably.

Queueing theory is the mathematical study of waiting lines, or queues (Sundarapandian, 2009). It analyzes key factors such as arrival rate, service rate, queue length, waiting time, and system capacity, all of which are critical in environments with limited resources or controlled access points. In public venues such as stadiums, transportation hubs, and sports complexes, queueing theory is used to optimize the design and operation of entry and exit flows, ticketing areas, and checkpoint management.

Fundamental Diagram of Pedestrian Flow (Helbing et al., 2007) further complements these frameworks by illustrating how pedestrian speed, density, and flow rate interact. The theory posits that, as density increases beyond a critical threshold, individual walking or jogging speeds decrease, and overall flow efficiency declines. This model provides a mechanism for the study to interpret how crowding levels at the oval affect mobility and comfort, especially when comparing observed peak-hour densities against theoretical capacity.

These theories offer a structured framework for evaluating the precision, responsiveness, and practical utility of people counting practices. Using these theoretical perspectives helps the study better assess whether CCSC's current systems deliver accurate data that fulfill the facility's operational requirements.

1.3. Literature Review

People Counter

Knowing how many people are in a room or building is essential for monitoring and security reasons. It aims for compliance with regulations, operational efficiency, and organizational success for individuals and businesses (Curiel et al., 2024). Whether for event coordination or maintaining optimal conditions, tracking occupancy is a fundamental aspect of efficiently managing any space. When the number of people exceeds a venue's intended capacity, it creates operational challenges, including congestion, inefficient use of resources, and difficulty accurately monitoring attendance. Traffic congestion and crowd-control challenges can significantly affect the efficiency of event operations and complicate data collection (Kopaczewski et al., 2023). Traffic congestion and crowd-control difficulties can further complicate events, leading to disruptions or postponements. However, traditional methods such as clickers, tally sheets, and manual hand counting are still used to count people today. Although these methods are effective for small crowds, larger audiences require more innovative and efficient solutions.

Effective crowd monitoring is essential for significant events. The operational challenges of managing high-volume traffic are highlighted by the frequent large-scale running events in the city, which require high-capacity counting and monitoring systems. The contrast between this vast volume and the CCSC's maximum daily attendance of 1,585 people in Table 1 indicates that manual counting is insufficient for effective crowd management in high-density athletic settings.

Statistical Tools in Questionnaire & Likert-Scale Analysis

Quantitative survey data, particularly those obtained from Likert-scale questionnaires, are often analyzed using descriptive statistics such as frequency distributions, percentages, means, and standard deviations. Frequency distributions organize responses into categories to show how frequently each occurs, while percentage distributions express the proportion of responses relative to the total number of participants. These techniques provide a clear picture of how responses are distributed across the scale and help identify dominant patterns (Yin et al., 2016).

Frequency distributions organize responses into categories to show how often each occurs, while percentages express the proportion of responses relative to the total number of participants. They are computed as:

$$P = \frac{f}{n} \times 100 \quad (1)$$

Where f is the frequency of a response category and n is the total number of respondents. These techniques provide a clear picture of how responses are distributed across the scale and help identify dominant patterns (Yin et al., 2016).

The mean is a measure of central tendency that represents the average response per item, calculated by dividing the sum of responses by the total number of respondents, with the formula:

$$\bar{X} = \frac{\sum X}{n} \quad (2)$$

Where $\bar{X}$ represents the mean, $\sum X$ is the total sum of all responses for a specific item, and n is the total number of respondents.

Meanwhile, the standard deviation quantifies variability by measuring how far responses deviate from the mean, with the formula:

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}} \quad (3)$$

Indicating whether opinions are consistent or widely dispersed (Yin et al., 2016). In Likert-based instruments, these two statistics are frequently used to examine the consistency and reliability of item responses.

To assess internal consistency, many studies employ Cronbach's Alpha. This coefficient measures how closely related items within the same construct are, with values ranging from 0 to 1, using the formula:

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum \sigma_i^2}{\sigma_t^2} \right) \quad (4)$$

Where k refers to the number of test items or statements under the same construct within the joggers' questionnaire, such as those grouped under "accuracy," "consistency," or "effectiveness" of the current people counting practices, σ_i^2 represents the variance of each item, and σ_t^2 denotes the variance of the total composite score across all items in that construct. A value of 0.70 or higher is generally considered

acceptable, signifying that the grouped items reliably measure the exact dimension of a concept (Jeong et al., 2023). Cronbach's Alpha is widely used in health, education, and social science research to validate instruments and confirm that the constructs being measured are internally reliable (Jeong et al., 2023).

Qualitative Analysis & Thematic Coding

For the qualitative component, thematic coding is a widely used method of analyzing interview and open-ended survey data. Thematic coding involves breaking down transcripts into smaller units of meaning, assigning codes to these segments, and grouping similar codes into overarching themes. These themes are then refined, compared, and interpreted to reveal patterns, perspectives, and recurring issues within the data (Vaughn & Turner, 2015).

The process often begins with open coding, in which initial codes are freely generated, followed by axial coding, in which codes are linked based on relationships or shared meanings. Finally, selective coding integrates the themes into a coherent narrative that addresses the research questions. To strengthen credibility, multiple coders may be involved to cross-check findings, and themes are continuously reviewed until saturation is reached, meaning no new insights emerge (Vaughn & Turner, 2015).

Quantitative methods such as descriptive statistics and Cronbach's Alpha, as well as qualitative analysis through thematic coding, have been frequently used and discussed in existing studies. Embedding both in this study ensures methodological

rigor and provides a balanced approach to understanding the effectiveness of people counting practices.

Fundamental Diagram of Pedestrian Flow

The Fundamental Diagram of Pedestrian Flow describes the quantitative relationship between pedestrian density, walking speed, and flow rate within a given space (Helbing et al., 2007). It explains how movement efficiency is affected by the number of people occupying the same pathway or lane. As density increases, individual speed tends to decrease, and beyond a critical threshold, overall flow begins to deteriorate due to physical interference among individuals. The speed-density, flow-density, and flow-speed diagrams characterize how people move under varying crowding conditions, illustrating this relationship through three interrelated curves.

Within the CCSC oval framework, this model helps explain how jogger movement patterns vary with different crowd levels. During low-density periods, joggers can move freely while maintaining a consistent speed and rhythm. However, as occupancy approaches the oval's lane capacity, congestion occurs, forcing individuals to adjust pace or shift lanes to avoid collisions. Studies by Zhang et al. (2019) and Daamen & Hoogendoorn (2020) highlight that even small increases in density can significantly reduce flow efficiency, particularly in circular tracks where continuous motion restricts lateral dispersal. This underscores the importance of monitoring occupancy levels to maintain both safety and performance.

Crowd Density Estimation Metrics

Crowd density estimation focuses on quantifying how many individuals occupy a defined area at any given time and assessing how this occupancy level influences mobility and safety. Fruin (1993) introduced a widely used standard that identifies four density zones:

- Free movement (<1.0 person/m²): individuals move freely with minimal interference.
- Restricted movement (1.0–2.0 persons/m²): minor trajectory adjustments occur.
- Crowded flow (2.0–4.0 persons/m²): individual speeds are limited by proximity.
- Critical crowding (>4.0 persons/m²): personal space disappears, and flow efficiency sharply declines.

Although these values were initially developed for pedestrian walkways, the same principles apply to jogging ovals, where spatial limitation per runner can affect rhythm, comfort, and safety. The concept of lane-based occupancy adapts Fruin's model to circular paths, where spacing between joggers determines whether movement remains fluid or becomes congested. For instance, Del Prete (2025) proposed that maintaining 1.8 to 2.2 meters per jogger per lane ensures continuous motion without forcing unnecessary overtaking or collisions.

Helbing and Molnár (1995) described crowd motion as a social force system in which each person adjusts their speed and trajectory to maintain personal distance while minimizing collisions. In dense jogging environments, this adjustment process can lead to “stop-and-go” waves, similar to traffic flow congestion. Recognizing these

thresholds is essential when comparing measured attendance data to theoretical lane capacities.

Finally, Haghani and Sarvi (2018) highlighted the importance of temporal granularity in density observation. Measuring occupancy at shorter intervals, such as every 10 minutes, captures transient surges that daily averages may hide. Applying this temporal resolution to the CCSC oval enables precise identification of peak periods, offering a more accurate understanding of jogger crowd dynamics without relying on automated detection systems.

While density metrics describe how tightly individuals occupy a space, understanding how they move within it requires examining human flow and motion patterns, which reflect both behavioral and spatial dynamics at varying occupancy levels.

Measuring occupancy helps determine whether the jogging oval operates within its safe and functional limits. This can be expressed quantitatively through the occupancy rate, defined as:

$$O = \frac{N_a}{N_c} \times 100 \quad (5)$$

Where O represents the occupancy rate (in percent), N_a is the actual number of joggers observed within the oval, and N_c is the calculated maximum capacity based on lane width and optimal spacing per jogger.

An occupancy rate approaching or exceeding 100% indicates congestion and reduced flow efficiency. This formula is consistent with facility utilization models used in sports facility research (Yano & Wakairo, 2006; Kang & Lee, 2022), which relate actual usage to the space's theoretical capacity to assess operational effectiveness.

While density metrics describe how tightly individuals occupy a space, understanding how they move within it requires examining human flow and motion patterns, which reflect both behavioral and spatial dynamics at varying occupancy levels.

Human Flow and Motion Pattern Analysis

Human flow analysis focuses on the quantitative description of how people move through shared spaces over time. When it comes to jogging tracks, human flow depends mainly on the number of joggers, their pace, and the available space per person. According to Still (2014), human flow follows principles similar to fluid dynamics. When density exceeds an optimal threshold, the speed of individual movement decreases, reducing overall flow efficiency. This concept is often described using the fundamental diagram of pedestrian movement, which relates flow (q), density (p), and velocity (v) through the relationship:

$$q = p * v \quad (6)$$

In open circular tracks such as the CCSC oval, flow analysis helps illustrate how joggers distribute themselves across lanes and time intervals. Daamen and

Hoogendoorn (2020) found that continuous-path users, such as runners on tracks, tend to form natural lane orderings, where differences in speed result in consistent spatial separation between faster and slower joggers. This structural distribution reduces lane interference and supports steady overall movement.

Temporal flow patterns, such as early-morning peaks or evening congestion, are also key to understanding jogger occupancy. Johansson et al. (2008) emphasized that movement dynamics are not random; they follow repeatable cycles tied to habitual routines and external conditions like temperature or lighting. For instance, morning sessions may exhibit steadier flow due to consistent runners, while evening sessions often exhibit higher variability due to mixed groups of walkers and joggers.

By analyzing flow and motion patterns in this manner, researchers can identify crucial density points at which movement shifts from free-flowing to congested, thereby providing a purely spatial and behavioral framework for assessing whether the CCSC oval operates within its optimal capacity.

People Counting with Manual Tally Sheets

Second Floor					
	individual	individual with computer	group	group with computer	TOTAL
PAC, Rm 245 - tables					
PAC, Rm 245 - 2 computers					
Study Rm 242A					
Study Rm 242B					
Reference Computers -					
Reference Rm - tables					
Mircoform Reader					
Reference Area Scanner/Printer Alcove					
Reference - 240A (Purple)					

Figure 1. Head-counting tally sheet by area and activity (Burel, 2016)

People Counting with Manual Tally Sheets refers to the traditional method of recording the number of people passing a point or entering an area by physically marking tallies on paper or using mechanical tally counters. The basic process involves an observer positioned at a strategic location who records each person by making vertical marks on a tally sheet, typically organizing them into groups of five, with the fifth mark drawn diagonally across the previous four to create easily countable bundles (Box, 2021).

The fundamental advantage of manual tally counting lies in its simplicity and accessibility. The method requires no technical equipment or infrastructure, making it immediately deployable in virtually any location (Affluences, 2023). This universality

means tally charts can be used regardless of language barriers, requiring only basic counting skills and the ability to make marks (Read, 2025). The low initial cost represents another significant benefit, as organizations need only provide paper, writing materials, and human observers rather than investing in expensive automated systems (V-Count, 2020).

However, manual tallying faces substantial limitations that significantly affect its reliability and practicality. Research consistently demonstrates that manual counting systematically underestimates actual pedestrian volumes, with error rates ranging from 8% to 25% compared to 2% to 5% for automated video counting systems (Diogenes, 2007). Dense pedestrian traffic exacerbates these issues, as observers struggle to maintain accurate counts when confronted with high-volume or rapidly moving crowds (Department of Traffic, 2016).

People Counting with Carry-on Radio Frequency Identification Tags

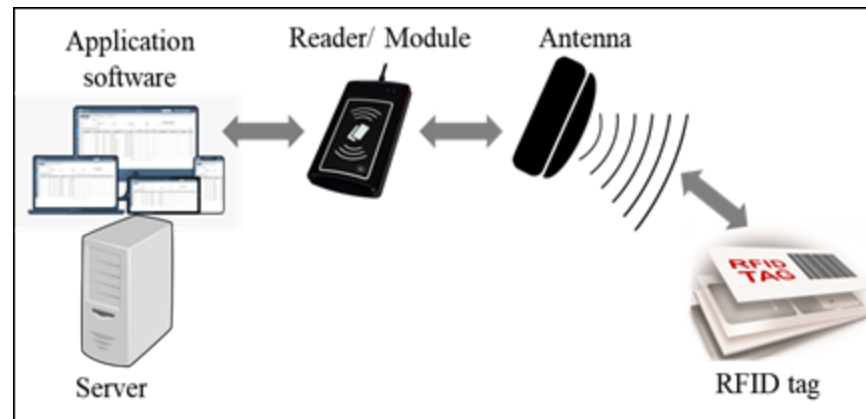


Figure 2. Basic radio frequency identification (RFID) system (Borowski, 2021)

The Radio Frequency Identification (RFID) tag is a technology that monitors the locations and movements of tagged individuals or objects by detecting radio waves emitted by the tag in real time when within the coverage zone. Munoz-Ausecha et al. (2021) conclude that RFID technology is widely used in various contexts, such as healthcare, logistics, asset tracking, and supply chain, all of which aid in monitoring presence and ensure accurate people counting and tracking. Wang et al (2023) proposed a new solution for people counting using carry-on RFID tags to enhance tracking accuracy and system reliability. Wang et al. (2023) also proposed that the system relies on reading profiles of RFID tags and uses both Hausdorff and Euclidean distance algorithms to achieve precise clustering.

According to Ma et al. (2020), the mobile RFID-based method offers high accuracy at reduced cost by using passive RFID. Ochi (2024) stated that RFID technology provides better privacy protection because it does not rely on facial recognition or

video-based monitoring, thereby addressing concerns about data privacy and surveillance. More than that, it also offers good performance in poor visibility conditions, as it is immune to lighting fluctuations, obstacles, and environmental limitations, which often undermine the accuracy of camera-based people-counting methods for applications such as retail analytics, queue management, and occupancy monitoring.

Although the RFID-based people-counting system has many advantages, it also has several disadvantages. One of the most significant drawbacks is its reliance on RFID tags, as individuals must carry RFID-tagged items to ensure accurate counting. Such may not always be practical to achieve for countless or inaccurately counted people, especially when some do not have the required tags. Furthermore, potential tag interference may affect system performance, as overlapping RFID tags or environmental factors such as metallic surfaces and electronic devices can disrupt readings (Almeida & Silva, 2021). The system also has a limited application scope and is not suitable for open public spaces or environments where people do not naturally carry RFID-tagged items. These challenges underscore the need to carefully consider the system's deployment context to ensure reliable operation.

RFID-based people counting is a valuable, privacy-preserving way to track people without using cameras. It works well in low to no lighting and is cheaper because it uses RFID tags (Ochi, 2024; Ma et al., 2020). However, it only counts people carrying these RFID tags, so that it might miss a few (Almeida & Silva, 2021). Along with signals, they sometimes get mixed up, making the system much less accurate (Wang et al.,

2023). Even with these issues, RFID remains a good choice for locations such as stores and hospitals (Munoz-Ausecha et al., 2021).

People Counting on Low-Resolution Infrared Arrays

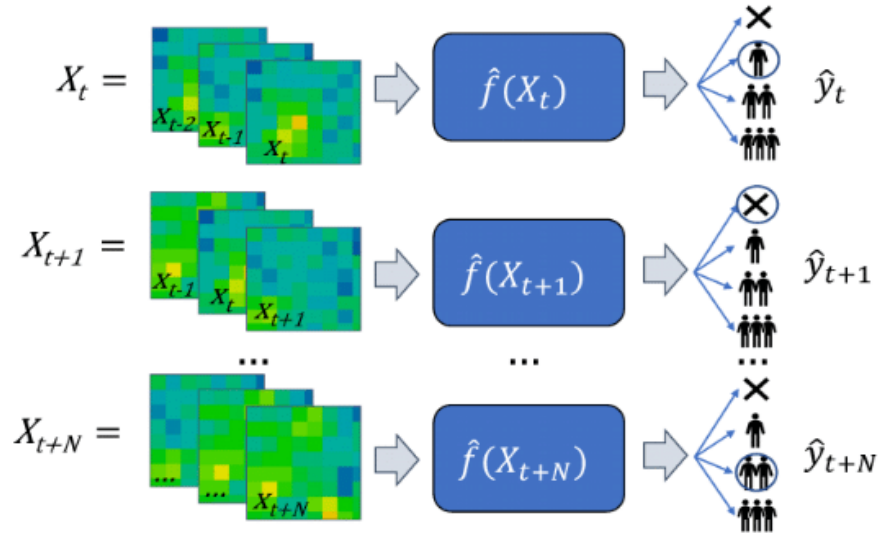


Figure 3. Low-resolution Infrared Arrays (Xie, 2023)

Infrared (IR) sensors are thermal-signal detectors that detect the heat emitted by human bodies, making them effective even in low-light or dark environments (Zhang et al., 2023). These devices are sensitive motion detectors to infrared radiation. When entering or exiting the sensor's field of view, a temperature change triggers a count. Xie et al. (2023) found that Deep Learning (DL) can achieve higher accuracy and efficiency than classic algorithms for people-counting tasks.

Low-resolution IR array sensors offer several advantages, making them an economical and efficient solution for people counting and activity recognition. According to Bouazizi et al. (2022), a 32x24-pixel IR sensor will deliver high accuracy at a cost

significantly lower than that of a traditional system. This clearly illustrates that leveraging low-end technology with advanced DL techniques provides economic advantages. All these sensors are energy-efficient, enabling real-time processing while consuming very low power and can withstand long-term use without frequent battery replacement, a crucial aspect of sustainable IoT applications (Newaz & Hanada, 2024). Their adaptability and flexibility enable their use not only for indoor positioning but also for human activity recognition, making them a versatile tool across many applications.

Despite their advantages, resolution IR array sensors have their own limitations. Low resolution might not be effective at distinguishing multiple individuals, particularly in crowded places, potentially degrading performance in certain conditions (Bouazizi et al., 2022). Moreover, according to Newaz & Hanada (2024), it is affected by environmental factors such as ambient temperature and obstructions, which may compromise data reliability and the sensor's overall performance. Another disadvantage is that these sensors rely on thermal signatures, and variations in clothing or other heat sources can cause inaccurate detection and make human recognition difficult in a variety of situations (Newaz & Hanada, 2024).

IR sensors count people by detecting their body heat, making them useful even in the dark (Zhang et al., 2023). They are cheap, use little power, and are good at tracking in real time (Bouazizi et al., 2022; Newaz & Hanada, 2024). Nevertheless, their limited resolution makes it difficult to count individuals when in congested areas, even when accuracy is impacted by temperature fluctuations or other heat sources (Bouazizi

et al., 2022). Even with these limitations, the IR sensor remains an excellent choice for homes and offices where privacy is crucial (Xie et al., 2023).

People Counting from Moving Camera

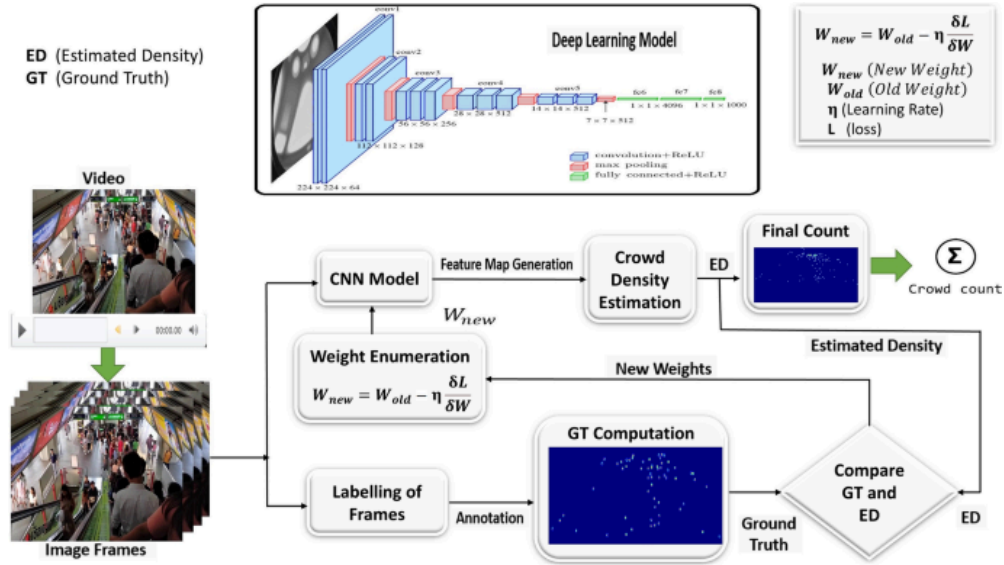


Figure 4. People Counting from Moving Camera Videos through PeopleNet Framework (Tomar, 2024)

This approach involves automatically detecting and counting people in video footage captured by cameras that are not fixed in place. Moving cameras can be handheld, mounted on drones, or mounted on cars, whereas stationary cameras are fixed in place. Moving cameras make counting people more challenging because of the subjects' distance, background, and camera location, which can vary frequently. These characteristics make it more challenging to observe and count people than with fixed-position cameras. In the most recent studies, Tomar et al. (2024) presented the PeopleNet framework, which uses a transfer-learning-based model to count people in videos recorded from a moving camera. This method saves time and money by using a

technique known as transfer learning, which enhances a model that has already been trained to assign new tasks.

People counting from a moving camera using DL techniques to improve crowd density estimation in dynamic environments. According to Moon et al. (2022), this would significantly enhance the precision of generated crowd density estimates and could be used to handle overlapping movements in complex situations. It also stated that it further diminishes training, as it can use only specific layers of VGG16 and fine-tune them efficiently, with strong feature extraction. It is more effective than traditional optical flow methods due to its robustness in handling scale variations and limited viewpoints (Moon et al., 2022).

However, this approach also has certain disadvantages. The reliance on hardware can be challenging because cloud implementations heavily rely on computational resources, making real-time deployment on edge devices less feasible. The system is sensitive to illumination changes, which can affect the accuracy outdoors, especially with rapid lighting changes (Zhang et al., 2020). Moreover, there is no public benchmark, making it rather impossible to compare the performance directly (Moon et al., 2022).

Although it has several drawbacks, this approach to counting people using a moving camera is helpful in many ways. As demonstrated by PeopleNet architecture, DL enhances accuracy, particularly in complicated situations (Tomar et al., 2024).

However, it depends on powerful hardware, making real-time use difficult on smaller devices. Accuracy can also be affected by changes in lighting, particularly outdoors (Zhang et al., 2020). Furthermore, researchers and developers cannot easily evaluate which method performs best in real-world settings without a standard benchmark or dataset (Moon et al., 2022).

People can be counted in a variety of ways, each with its own pros and cons. RFID tags track people without cameras, offering privacy and working in dark places, but they require people to carry tags, and signals can interfere (Och, 2024; Wang et al., 2023). IR sensors, although effective in low light and energy-efficient because they can sense body heat, struggle in crowded spaces and near other heat sources (Bouazizi et al., 2022; Newaz & Hanad, 2024). Moving cameras use DL to count people in videos, improving the accuracy in challenging scenarios, but they require strong hardware and are affected by lighting changes (Tomar et al., 2024; Moon et al., 2022). The best method depends on the specific environmental needs.

THE PROBLEM

1.4. Statement of the Problem

The researchers conducted this study to assess current people-counting practices at the Cebu City Sports Complex. Specifically, this study sought to answer the following questions:

1. What specific time periods can be identified as peak periods of jogger turnout at the CCSC oval, based on observed patterns of use and overall turnout distribution throughout the day?
2. What challenges do CCSC staff encounter in implementing and maintaining the current people counting for joggers?
3. How do joggers perceive the facility's ability to monitor and manage crowd levels at the CCSC oval, particularly in maintaining safe and comfortable jogging conditions?
4. Based on the findings of the study, what recommendations can be proposed to strengthen and enhance the current people-counting system at the CCSC oval for improved people-counting efficiency and accuracy?

1.5. Significance of the Study

The current study is crucial, as the increasing number of joggers and recreational users at the Cebu City Sports Complex (CCSC) poses challenges in monitoring attendance, ensuring safety, and effectively managing facility usage. As highlighted by the daily volume of joggers and the large-scale running events in Cebu City, reliance on traditional manual counting methods, such as tally sheets and clickers, has become prone to errors, delays, and inefficiencies. These issues often result in inaccurate records, difficulty in managing crowd flow, and missed opportunities for data-driven planning. This research aims to address these operational gaps by assessing people-counting practices, exploring crowd-monitoring practices, and formulating proactive measures to improve safety, user experience, and facility monitoring.

This study is going to be beneficial to the following:

CCSC Administrators and Venue Managers: This study will provide them with accurate, data-driven strategies to monitor attendance, allocate resources more efficiently, and enhance decision-making for scheduling and jogger monitoring at the oval (Curiel et al., 2024).

Security Personnel and Maintenance Staff: The research output will equip them with tools and insights to perform real-time monitoring, anticipate crowding issues, and

respond effectively to irregularities or emergencies, thereby ensuring smooth and safe operations within the sports complex (Kopaczewski et al., 2023).

Local Government and Sports Development Officials: The study's findings can inform the development of better guidelines, regulations, and policies for managing public sports facilities. Reliable attendance data will also support future investments in infrastructure, training programs, and athletic event planning (Kadi et al., 2024).

Joggers and Regular Facility Users: By adopting efficient monitoring systems, joggers will benefit from improved safety, reduced congestion, and a more enjoyable experience while using the CCSC oval. Their feedback on crowding and convenience will also directly shape future improvements to the facility (Papalexi & Bamford, 2020).

Technology Developers and Researchers: This study provides practical insights into applying people-counting systems in open-air, high-traffic environments. It highlights challenges such as weather exposure, multiple entry points, and varying user flow, offering valuable lessons for further research and innovation in monitoring technologies (Al-Qurashi et al., 2023).

By promoting accurate people counting and real-time monitoring practices, the study contributes not only to more efficient monitoring of the CCSC athletic oval but also to the broader goal of improving public safety and recreational infrastructure in Cebu

City. The integration of theoretical perspectives such as Occupancy Density Theory and Queueing Theory will provide the framework for analyzing spatial usage, density-based analytics, and entry–exit flow monitoring. Understanding jogger behavior, movement patterns, and space utilization is essential for designing effective monitoring systems and ensuring sustainable, safe, and enjoyable use of athletic facilities.

FLOW OF RESEARCH

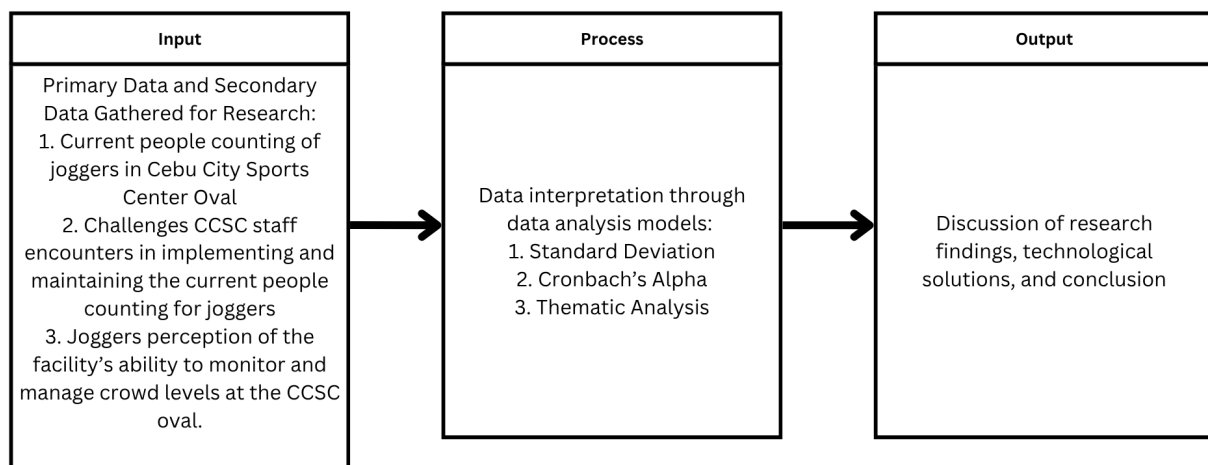


Figure 5. Research Flow Diagram

The study begins by collecting both primary and secondary data, focusing on three key areas: the current people counting system used for joggers at the Cebu City Sports Center (CCSC) Oval, the challenges encountered by CCSC staff in implementing and maintaining this system, and the joggers' perception of the facility's ability to monitor and manage crowd levels. The collected data are then analyzed using various data interpretation models, including Standard Deviation to measure variability and consistency in responses, Cronbach's Alpha to assess the reliability of the survey instruments, and Thematic Analysis to extract insights from qualitative feedback. The

results of these analyses lead to a detailed discussion of the research findings, the identification of technological gaps, and the formulation of practical solutions aimed at enhancing real-time people counting and overall crowd monitoring efficiency at the CCSC Oval.

RESEARCH METHODOLOGY

1.6. Research Design

This study adopts a mixed-methods research design combining quantitative and qualitative approaches to evaluate current people-counting practices for joggers at the CCSC oval. It aims to determine the accuracy, consistency, and operational efficiency of the manual counting methods currently used by CCSC personnel.

The research will be conducted within the CCSC oval, focusing on crowd monitoring during peak and non-peak hours. Respondents include 8 staff members involved in ticketing, security, and visitor documentation, and 65 joggers, a sample size statistically derived from the Average Daily Attendance ($N = 1220$) as recorded in Table 11. Calculated using a 90% confidence level and a 10% margin of error, this sample size was selected to accurately represent the crowd volume of a standard operational day while ensuring the data collection process remained feasible for the researchers.

Data will be gathered through systematic observation, structured interviews, and survey questionnaires. Observations will record real-time jogger counts using the “Click Counter” mobile application, while interviews and surveys will capture staff experiences and jogger perceptions of crowding and facility use. A preliminary dry run will be conducted to ensure clarity and the feasibility of the instruments.

Collected data will undergo quantitative analysis using frequency, percentage, mean, and standard deviation, while Cronbach's Alpha will assess internal reliability. Qualitative data from interviews will be analyzed through thematic coding to identify recurring operational issues.

This design provides a comprehensive framework for understanding and improving the accuracy and consistency of people-counting practices at the CCSC oval.

1.7. Research Environment

The research will be conducted at the Cebu City Sports Complex (CCSC), specifically within the athletic oval, which is one of the most frequented areas for jogging, exercise, and athletic training (Magsumbol & Mascardo, 2025). The oval is an open-air track located within the sports complex along Osmeña Boulevard in Cebu City. It is approximately 398.5 meters in length, slightly shorter than the standard 400 meters due to non-compliant lane markings applied during recent renovations (Seblos, 2024). The track consists of eight running lanes, each about 1.22 meters wide, which conforms to international standards for athletics track lane widths (Del Prete, 2025; World Athletics Technical Rules, 2024). The oval also encloses an inner grassy field used for other physical activities. The entire track is enclosed by a concrete barrier and stadium seating with three main access points: primary entrances/exits on the northern and southern sides, plus a secondary entrance on the eastern boundary, typically opened during peak foot traffic hours.



Figure 6. Cebu City Track Oval. CDN News

Capacity estimates based on these measurements and an average spacing of approximately 2 meters per jogger along the track length suggest a maximum theoretical occupancy of roughly 200 joggers per lane and up to 1,600 joggers concurrently using all eight lanes (Del Prete, 2025). The basic capacity calculation per lane is:

$$\text{Capacity per lane} = \frac{\text{Track Length}}{\text{Space per Jogger}} = \frac{398.5 \text{ meters}}{2 \text{ meters per jogger}} \approx 200 \text{ joggers}$$

However, jogging speed varies by lane due to biomechanical factors associated with the curve radius, with slower speeds in the inside lanes and faster speeds in the outer lanes. Adjusted spacing accounts for this speed difference, with inner lanes estimated at 1.8 meters per jogger and outer lanes at up to 2.2 meters per jogger (Del Prete, 2025). This results in an adjusted approximate capacity per lane as follows:

Table 2. Estimated Jogger Capacity per Lane

Lane Type	Lane #	Estimated Space per Jogger (meters)	Estimated Capacity per Lane (joggers)
Inner	1	2.2	181
Inner	2	2.2	181
Middle	3	2.0	199
Middle	4	2.0	199
Middle	5	2.0	199
Middle	6	2.0	199
Outer	7	1.8	222
Outer	8	1.8	222
Total			1,602

Multiplying the lane-wise capacity by the number of lanes yields a theoretical maximum capacity of approximately 1,600 joggers across all eight lanes. Fieldwork will collect real-time jogger density and flow data to validate and refine these capacity estimates, helping to identify peak congestion and access-point pressures (Del Prete, 2025).

The CCSC athletic oval presents a semi-controlled, high-traffic environment ideal for evaluating the performance of people-counting practices. Variables such as uneven foot traffic patterns, exposure to weather, and multiple access points provide realistic challenges for automated counting systems (Curiel et al., 2024; Kopaczewski et al., 2023). The oval’s current reliance on manual counting methods underscores the need

for modernization, particularly during high-traffic periods when real-time monitoring is essential to maintaining operational efficiency and accurate occupancy tracking (Magsumbol & Mascardo, 2025).

1.8. Research Respondents

The respondents for this study include the specialized personnel responsible for on-ground monitoring and the active users of the Cebu City Sports Center (CCSC) oval.

Track Marshals serve as the primary operational respondents. A total of four (4) Track Marshals are purposively selected as they are the specific personnel tasked with the direct, real-time monitoring of the oval track. These marshals are stationed within the track area, providing them with continuous observation of physical density and operational conditions. Their input is vital for identifying the limitations of current manual practices and validating the data blind spots occurring on the field.

The Joggers who utilize the facility during peak operating times serve as the user-level study participants. The sample size for this group is determined by analyzing preliminary data for each specific session. The researchers calculated the average attendance for the morning and evening sessions separately to ensure each period is accurately represented. Specifically, morning joggers are identified as those who utilize the facility during the 5:00 AM to 8:00 AM window, while evening joggers are those who utilize the facility between 5:00 PM and 8:00 PM. By applying a Sample Size Calculator with a 90% Confidence Level and a 10% Margin of Error to these averages, the study identifies a requirement of 56 morning joggers and 60 evening joggers, totaling 116

respondents. This approach allows the researchers to investigate how occupancy levels are perceived and how the facility performs during these distinct periods of operation.

This sampling strategy is selected over utilizing the total weekly aggregate to address the recurrence of the facility's user base. Since the CCSC Oval functions as a routine-based fitness hub where specific individuals visit multiple times a week, using a cumulative weekly total would inflate the population size with redundant counts of the same individuals. Therefore, the Average Daily Attendance provides the most accurate representation of the unique population at any given time, ensuring the study remains operationally feasible while capturing a statistically significant snapshot of user perception across both morning and evening sessions.

Collectively, the four Track Marshals serve as an exclusive form of operational insights, which would serve as an added value to the results obtained from the 116 joggers. By focusing on the on-ground personnel, the study would ensure that the analysis of the people-counting and crowd-monitoring approaches is based on the actual conditions, which would serve as an adequate foundation for the proposed automated system.

1.9. Research Instruments

This study employs a mixed-methods approach that incorporates systematic observation, structured interviews, and survey questionnaires to collect data on crowd monitoring at the Cebu City Sports Complex (CCSC) oval.

Systematic Observation and Video Monitoring involves recording the number of persons entering and exiting the CCSC oval at set intervals, especially during peak usage times. To ensure the acquisition of precise, objective usage data, the study employed a non-intrusive video surveillance strategy. This process utilized a dual-camera setup to provide a basis for validating current manual crowd monitoring methods and identifying discrepancies between observed counts and reported figures.



Figure 7. Blurams A31 Camera

Figure 7 shows the Blurams A31 surveillance unit used to capture wide-angle footage of the oval. Researchers reviewed the video recordings from this device to perform manual headcounts of joggers utilizing the track. This equipment provided the

visual data necessary to quantify the total number of users present during each hourly interval throughout the data collection period.



Figure 8. V720 (A9 mini WiFi) Camera

Figure 8 depicts the V720 (A9 mini WiFi) camera system employed for localized monitoring at specific access points. The footage retrieved from this device was utilized by the researchers to manually count joggers entering and exiting the facility hour by hour, ensuring that flow patterns were accurately recorded and cross-referenced with the general oval counts.

The cameras were strategically positioned at the oval's access points on the northern and southern sides, which function as the primary corridors for jogger entry and exit. This location strategy was essential for the data collection process, as it provided a clear, unobstructed line of sight that allowed the researchers to manually track and quantify the volume of users entering the facility from either direction,

ensuring that the hourly headcount analysis was based on complete and verified visual evidence.



Figure 9. Video footage from the V720 (A9 mini WiFi) Camera, capturing the northern entrance of the Oval.

Figure 9 presents a sample frame from the V720 camera positioned at the northern entrance. This footage served as a primary source for the study's data analysis, allowing researchers to visually identify and manually tally every jogger entering through this gate in hourly intervals to establish accurate daily turnout volume.



Figure 10. Video footage from the Blurams A31 camera, capturing the southern entrance of the Oval.

Figure 10 displays the wide-angle field of view captured by the Blurams A31 camera near the southern entrance. Researchers used this footage to conduct manual hourly headcounts of the active joggers on the track. This visual record enabled the team to verify crowd density and determine the specific peak periods of facility usage based on actual observed headcounts rather than estimates.

Structured Interviews will be conducted with staff involved in crowd monitoring, including ticket booth attendants, security personnel, and visitor log managers. A standardized list of questions will be used to capture consistent data on their current manual crowd-counting practices, challenges encountered, and operational procedures followed during events.

Survey Questionnaires distributed to CCSC visitors with jogger passes will use closed-ended and Likert-scale questions to gather data on perceptions of crowd sizes, waiting times at entry points, and experiences with current crowd-monitoring operations. This allows assessment of the end-user's perspective on the effectiveness of manual counting and crowd management.

Together, triangulating systematic observation data, staff interviews, and visitor surveys will enable a comprehensive understanding of the accuracy and practices of crowd monitoring at CCSC.

1.10. Dry-Run Procedure

A preliminary dry-run will be conducted to ensure that all research instruments and procedures are feasible, reliable, and appropriate for actual data collection. This activity will be held at the Cebu City Sports Complex (CCSC) athletic oval during non-peak hours to minimize disruptions to both facility operations and joggers. The dry-run will involve orienting a small group of respondents, including two staff members and five to ten volunteer joggers, about the objectives of the study and their role in the testing process. For the interviews, personnel from the ticketing booth and jogger monitoring will be asked sample questions to determine the clarity and flow of the guide. Systematic observation will be tested by recording jogger entries and exits over a thirty-minute window using the "Click Counter" mobile application to verify the practicality of observation protocols. Likewise, survey questionnaires will be distributed to a small group of joggers with jogger's passes to check for clarity, comprehension, and completion time. Feedback from participants and observers will then be collected and

used to refine the instruments. The dry run will also serve as an evaluation of logistics, including access to respondents, positioning of observers, coordination among research assistants, and the time required to complete each method. Adjustments will be made accordingly to ensure that the main data collection runs smoothly and effectively.

1.11. Data Collection

The data collection for this study will be executed through a triangulated approach to ensure that objective headcounts align with the experiences of both the staff and the facility users. This process will begin with systematic observation using a dual-camera setup positioned at the northern and southern access points of the oval. Video monitoring will be conducted during the primary operational windows, specifically the Morning Session (5:00 AM – 8:00 AM) and the Evening Session (5:00 PM – 8:00 PM). Researchers will review this footage to perform manual headcounts in hourly intervals, establishing a baseline for track density and identifying the specific periods when the facility reaches maximum capacity.

In addition to the visual data, there will also be structured interviews with the Track Marshals who are in charge of ground-level monitoring. This will also provide insight into the qualitative problems faced in the process of counting during the identified peak hours. Finally, structured questionnaires will also be administered to the joggers using the facility during the identified peak sessions. This will help the researchers identify if the statistical peaks observed during the study correlate with the perceived user experience in terms of the availability of space.

1.12. Data Treatment

The data gathered in this study will be analyzed using both quantitative and qualitative methods, in line with the mixed-methods research design. Manual counts of jogger turnout will be summarized using frequency distributions and percentages across time intervals. Line graphs will be generated to illustrate daily turnout trends and to highlight peak-hour patterns. To determine whether jogger turnout exceeds the CCSC oval's carrying capacity, occupancy rates will be calculated and compared against the theoretical lane-based maximum capacity. The estimated space per jogger was set at 1.8 meters for the inner lanes (capacity $\approx$ 181 joggers), 2.0 meters for the middle lanes (capacity $\approx$ 199 joggers), and 2.2 meters for the outer lanes (capacity $\approx$ 222 joggers). Multiplying these lane-wise values by the number of lanes produces a total theoretical maximum capacity of approximately 1,600 joggers across all eight lanes. To determine the level of oval utilization, the observed jogger turnout will be compared against the estimated maximum capacity using Eq. (5):

$$O = \frac{N_a}{N_c} \times 100$$

Where O represents the occupancy rate (in percent), N_a is the actual number of joggers observed within the oval, and N_c is the calculated maximum capacity based on lane width and optimal spacing per jogger, an occupancy rate exceeding 100% indicates overcrowding.

For the quantitative component, data gathered from the jogger questionnaires will be encoded, cleaned, and analyzed using descriptive statistics. Frequency distributions

and percentage calculations will be employed to organize and summarize responses across the Likert-scale categories. These tools will be applied using the formulas for frequency (f) and percentage, using Eq. (1):

$$P = \frac{f}{n} \times 100$$

Where f is the frequency of a response category and n is the total number of respondents. These techniques provide a clear picture of how responses are spread across the scale and help in identifying dominant patterns (Yin et al., 2016).

To further analyze central tendency and variability, the mean, using the formula of Eq. (2):

$$\bar{X} = \frac{\sum X}{n}$$

Where X represents the mean, $\sum X$ is the total sum of all responses for a specific item, and n is the total number of respondents, and the standard deviation, using the formula of Eq. (3):

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

Will be computed. As discussed in the RRL (Yin et al., 2016), these measures provide insights into the average response per item and the dispersion of responses, thereby showing whether joggers' assessments are consistent or dispersed.

To ensure internal consistency of grouped questionnaire items under the same construct, Cronbach's Alpha will be computed using the formula of Eq. (4):

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum \sigma_i^2}{\sigma_t^2} \right)$$

Where k refers to the number of test items or statements under the same construct within the joggers' questionnaire, such as those grouped under "accuracy," "consistency," or "effectiveness" of the current people counting practices, σ_i^2 represents the variance of each individual item, and σ_t^2 denotes the variance of the total composite score across all items in that construct. This reliability test, with values of 0.70 or higher, is considered acceptable and will determine whether the items effectively measure the same underlying concept, following the standards cited in prior studies (Jeong et al., 2023).

For the qualitative component, data obtained from interviews with CCSC personnel, particularly those involved in jogger monitoring, documentation, and ticketing, will be transcribed and analyzed through thematic coding. This process will follow the framework described in the RRL (Vaughn & Turner, 2015), beginning with open coding to generate initial categories, axial coding to cluster related codes, and selective coding to integrate overarching themes. This method ensures systematic identification of recurring challenges and practices while enhancing the credibility of results through coder cross-checking and theme saturation.

By integrating these quantitative and qualitative procedures and grounding them in the methodological tools discussed in the Review of Related Literature, the study ensures both statistical rigor and contextual depth in evaluating the effectiveness of current people-counting practices.

CHAPTER 2

PRESENTATION, INTERPRETATION, AND ANALYSIS OF DATA

This chapter presents, analyzes, and interprets the data collected to assess the current people-counting practices at the Cebu City Sports Center (CCSC) Oval. The findings are derived from a mixed-method research design, integrating quantitative data from structured surveys and objective video-based headcounts, alongside qualitative insights from staff interviews.

The data presentation is organized sequentially according to the specific problems raised in the study. First, it establishes the peak periods of jogger turnout through a triangulation of camera-based observational data and self-reported jogger schedules. Second, it explores the challenges encountered by CCSC staff in maintaining manual counting procedures. Third, it evaluates the joggers' perception regarding the facility's ability to monitor crowd levels and maintain safety. Finally, the chapter presents proposed recommendations and the willingness of stakeholders to adopt technological enhancements for crowd management.

To ensure comprehensive data representation and address previous panel recommendations, the study engaged a total of one hundred twenty-four (120) respondents, consisting of one hundred sixteen (116) joggers and four(4) CCSC staff members. The jogger sample was rigorously stratified to represent the facility's bimodal usage, surveying 56 morning joggers and 60 evening joggers. This ensures the data

captures a balanced cross-section of users during both distinct peak periods. The 4 staff members comprise the total population of personnel directly involved in frontline jogger monitoring.

2.1 Identifying Peak Jogger Turnout Periods at the CCSC Oval

This section addresses the first statement of the problem regarding the identification of specific time periods characterized by peak jogger turnout. To ensure the accuracy of the findings, the study employed a triangulation method, integrating objective data from one week of camera-based headcounts with the subjective self-reported schedules of regular joggers, and staff observation. The following data presents the temporal distribution of facility usage, highlighting the specific days of the week and time intervals where the volume of users approaches or exceeds the oval's theoretical capacity.

2.1.1 Operational Definition of Track Congestion and Occupancy Thresholds

To establish a verifiable and scientifically grounded baseline for this study, congestion is objectively defined using the facility's spatial carrying capacity rather than subjective visual estimates. The evaluation of crowd density is anchored on the principles of Pedestrian Level of Service (LOS) (Fruin, 1971), a standard framework utilized in traffic engineering and facility management to classify the quality of movement within a confined space.

By adapting this framework to athletic facilities, the study categorizes the track's operational state based on its Occupancy Rate. Based on the capacity estimations established in Table 2, the primary running lanes (Lanes 1 and 2) possess a Functional Capacity (\$N_c\$) of approximately 181 joggers, maintaining a safe trailing distance of 2.2 meters per runner.

To systematically classify the manual headcount data, the study utilizes the following utilization thresholds:

Table 3. Occupancy Rate Thresholds and Level of Service (LOS)

Occupancy Rate (O)	Level of Service (LOS)	Operational Classification	Description of Track Conditions
0% – 49%	LOS A - B	Low Utilization (Free Flow)	Joggers have complete freedom to select their desired speed and running path. No overtaking conflicts.
50% – 79%	LOS C	Moderate Utilization (Stable Flow)	Normal running speeds can be maintained, but overtaking requires minor trajectory adjustments.
80% – 99%	LOS D	High Utilization (Restricted Flow)	Nearing functional capacity. Freedom of movement is noticeably restricted; the probability of conflict during overtaking increases.
100% and above	LOS E - F	Congestion (Overcrowded /	Functional capacity is breached. The

Forced Flow)

2.2-meter safety cushion is violated, forcing joggers to reduce pace or execute forced, unsafe lane changes.

Therefore, Functional Congestion is operationally defined in this study as any period where the track's utilization reaches LOS E - F (100% and above). At this critical density, the physical space available per user is statistically insufficient to maintain standard running mechanics, converting the track from a conducive training environment into an operational hazard.

2.1.2 Verification and Data Analysis Protocol

The video footage captured from November 29, 2025, to December 5, 2025, underwent a rigorous and comprehensive manual review process. To quantify the precise volume of users occupying the track, the researchers conducted systematic headcounts at hourly intervals throughout the entire seven-day observation period.

Crucially, a significant discrepancy exists between the facility's overall ticketing ingress and the actual track occupancy. To systematically resolve the number of people inside the facility who were not utilizing the track and field (e.g., infield athletes, Zumba participants, or grandstand spectators), the manual video review process employed strict spatial delineation and activity filtering. The researchers were explicitly instructed to tally only those individuals who were physically located within the boundaries of the eight-lane rubberized track and actively engaged in continuous walking or jogging.

Anyone occupying the grassy infield, seated in the grandstands, standing stationary, or moving perpendicularly across the track to access other areas was systematically excluded from the headcount (N_a). This strict observational criteria ensured that the data accurately measured the operational strain on the running track itself, rather than general facility attendance."

To ensure the reliability of this count and eliminate the potential for human error inherent in manual tallying, the study implemented a robust cross-verification protocol involving a rotating pair system among the three researchers. For every day of the observation period, two researchers were assigned to independently review and count the jogger turnout from the same video footage. This dual-counting assignment followed a specific rotational schedule to ensure balanced verification and objectivity.

Table 4. Daily Researcher Assignment for Cross-Verification of Video Counts

Observation Date	Assigned Counter A	Assigned Counter B
Saturday	Researcher 1	Researcher 2
Sunday	Researcher 1	Researcher 3
Monday	Researcher 2	Researcher 3
Tuesday	Researcher 1	Researcher 2
Wednesday	Researcher 1	Researcher 3

Thursday	Researcher 2	Researcher 3
Friday	Researcher 1	Researcher 2

By having two independent counters for every hourly segment, the researchers were able to cross-reference their tallies. Any discrepancies between the two counts were reviewed to establish a final, verified headcount, ensuring that the data presented in this study is accurate and free from individual observer bias.

Following the rigorous verification process described in the previous section, the final consolidated dataset is presented below. To establish a more comprehensive baseline and address the need for longitudinal analysis, the initial hourly jogger turnout recorded from November 29, 2025, to December 5, 2025, was expanded to include additional observation data recorded from February 20, 2026, to March 2, 2026. Table 4 details this aggregated hourly turnout across the expanded observation period. This updated tabulation represents the facility usage, providing a granular view of user volume that highlights recurring congestion windows and daily fluctuations that are otherwise invisible in daily aggregate totals.

Table 5. Jogger turnout by day and time on November 29, 2025 to December 5, 2025

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
5:00 AM	3	2	1	2	3	0	2
6:00 AM	66	5	3	38	29	29	39
7:00 AM	103	149	44	135	140	115	75
8:00 AM	81	212	35	158	165	123	60
9:00 AM	54	75	31	70	72	84	35
10:00 AM	4	17	27	14	8	18	20
11:00 AM	1	29	19	16	18	13	4
12:00 PM	1	1	2	3	3	4	32
1:00 PM	3	4	1	4	5	3	1
2:00 PM	0	2	1	1	1	0	0
3:00 PM	0	1	0	0	1	1	1
4:00 PM	41	37	1	42	39	49	40
5:00 PM	63	77	72	75	67	88	33
6:00 PM	85	103	95	98	105	113	55
7:00 PM	109	166	150	155	94	170	135
8:00 PM	217	229	220	223	217	107	156
9:00 PM	209	213	200	205	202	54	178

The table illustrates the jogger turnout by time and day for the week of November 29, 2025, to December 5, 2025. The data reveals two distinct peak periods for jogging, with high activity observed in the mornings around 7:00 AM to 8:00 AM and in the

evenings between 7:00 PM and 9:00 PM. Tuesday recorded the highest volume of joggers with a total of 1,322, followed closely by Thursday, while the single busiest hour of the week occurred at 8:00 PM. Conversely, the mid-day hours from 10:00 AM to 3:00 PM consistently showed the lowest turnout across all days.

Table 6. Jogger turnout by day and time on February 20, 2026 to February 26, 2026

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
5:00 AM	3	0	1	0	4	6	16
6:00 AM	5	1	3	0	22	32	71
7:00 AM	45	26	48	78	138	120	135
8:00 AM	43	115	38	159	128	171	159
9:00 AM	50	80	29	65	75	88	48
10:00 AM	6	15	24	6	5	42	52
11:00 AM	2	15	21	14	15	10	5
12:00 PM	0	2	1	2	2	5	28
1:00 PM	2	0	2	3	0	2	2
2:00 PM	0	1	0	2	0	0	0
3:00 PM	0	0	1	0	1	0	2
4:00 PM	4	0	2	13	42	51	33
5:00 PM	61	82	68	71	64	92	35
6:00 PM	89	108	91	102	110	142	160
7:00 PM	115	154	160	98	171	183	142
8:00 PM	111	104	118	118	217	220	231
9:00 PM	95	109	105	110	198	206	172

Table 6 details the hourly jogger turnout recorded during the expanded observation period, revealing a pronounced bimodal utilization pattern. This is

characterized by a significant morning surge, peaking at 171 joggers on Saturday at 8:00 AM, and a critical evening influx that consistently reaches its maximum density between 8:00 PM and 9:00 PM. Unlike earlier observations, this specific data set highlights a substantial escalation in weekend congestion, with the facility's highest volume recorded on Sunday at 8:00 PM (231 joggers), followed closely by Saturday evening (220 joggers) and Friday evening (217 joggers).

Crucially, the data indicates that the facility frequently approached or breached its Functional Inner-Lane Capacity of 181 joggers during the latter half of the week. Specifically, the capacity was exceeded during the evening rushes on Friday (217 joggers), Saturday (220 joggers), and Sunday (231 joggers). In contrast, Monday through Wednesday remained within the operational safety limits, despite a notable Tuesday morning spike of 115 joggers.

The presence of these high-volume weekend peaks justifies the methodological expansion to a 14-day longitudinal analysis. By capturing these weekend surges, the study avoids the risk of underreporting facility strain that would occur if only weekday data were considered. This comprehensive tracking establishes a stable and reliable empirical baseline for the facility's peak congestion levels, providing the necessary evidence for the implementation of an automated counting system.

Table 7. Jogger turnout by day and time on February 27, 2026 to March 5, 2026

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
5:00 AM	1	0	0	2	14	25	32
6:00 AM	45	3	5	2	32	76	103
7:00 AM	66	17	30	42	155	195	210
8:00 AM	111	125	155	132	185	170	185
9:00 AM	65	105	80	95	85	120	135
10:00 AM	28	34	22	30	25	55	60
11:00 AM	10	3	8	6	8	40	45
12:00 PM	2	0	0	0	0	2	11
1:00 PM	0	0	0	0	4	1	0
2:00 PM	1	3	0	1	4	11	4
3:00 PM	0	2	2	1	1	16	20
4:00 PM	5	6	2	16	11	22	21
5:00 PM	87	95	85	110	130	165	180
6:00 PM	155	145	120	165	185	220	235
7:00 PM	115	105	190	130	210	275	233
8:00 PM	195	155	140	165	245	281	271
9:00 PM	135	120	105	115	185	240	265

Table 7 illustrates the hourly jogger turnout for the final observation period, capturing a significant escalation in facility utilization and an extreme weekend surge. While the established bimodal pattern persists featuring a morning peak of 210 joggers on Sunday at 7:00 AM and a severe midday trough the evening data reveals a critical escalation in the facility's operational load during high-traffic intervals.

During this observation week, the Functional Inner-Lane Capacity limit of 181 joggers was breached on four out of the seven days. While mid-week figures for Tuesday, Wednesday, and Thursday remained within the capacity threshold, Monday evening recorded an early surge of 215 joggers at 7:00 PM. The most significant saturation occurred during the weekend window, with Friday, Saturday, and Sunday all consistently exceeding the 181-jogger limit. This period culminated in a critical bottleneck of 281 joggers on Saturday at 8:00 PM and 271 joggers on Sunday at 8:00 PM.

This massive, sustained influx during the weekend underscores the volatile nature of the facility's utilization. It further validates the necessity of a three-week longitudinal analysis to synthesize these weekly fluctuations into a stable empirical baseline. By documenting these extreme surges, the researchers have confirmed that manual counting practices are insufficient for managing the current levels of facility congestion, particularly when volumes exceed the carrying capacity by nearly 55% during peak weekend hours.

2.1.3 Analysis of Jogger Turnout Patterns

To synthesize the diverse temporal patterns observed across the individual weeks and establish a definitive operational baseline, the data from the three observation periods (Table 5, Table 6, and Table 7) were aggregated and averaged. While the isolated weekly datasets were critical for identifying specific anomalies, such as the massive weekend surges observed in late February and early March, calculating

a longitudinal average is methodologically essential. This aggregation smooths out temporary irregularities caused by localized events, shifting weather conditions, or spontaneous group activities. By calculating the mean hourly turnout across the full 21-day dataset, the researchers derived a stabilized, reliable representation of the facility's true usage dynamics. Table 8 presents this consolidated Three-Week Average Jogger Turnout, which serves as the primary dataset for evaluating the track's consistent bimodal peak periods and mathematically verifying the frequency of functional capacity breaches.

Table 8. Three-Week Average Jogger Turnout Patterns

Time	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
5:00 AM	6	6	4	7	7	10	13
6:00 AM	58	30	25	44	54	67	76
7:00 AM	98	115	74	126	144	143	119
8:00 AM	78	181	76	168	155	140	108
9:00 AM	56	87	47	77	77	97	73
10:00 AM	13	22	24	17	13	38	44
11:00 AM	4	16	16	12	14	21	18
12:00 PM	1	4	3	3	3	9	24
1:00 PM	2	3	4	3	4	9	10
2:00 PM	0	2	4	1	2	4	7
3:00 PM	1	1	1	0	1	15	18
4:00 PM	37	31	14	38	52	65	61
5:00 PM	80	85	75	85	87	115	83
6:00 PM	110	119	102	122	133	158	150
7:00 PM	146	181	165	182	134	209	189
8:00 PM	188	218	188	202	226	207	234
9:00 PM	166	181	170	177	195	167	205

Table 8 presents the comprehensive Three-Week Average Jogger Turnout, synthesized from observations across November 29, 2025 to December 5, 2025 and February 20, 2026 to March 5, 2026. This consolidated dataset effectively smooths out localized anomalies, such as isolated weekend surges or specific weekday weather disruptions, to establish a highly stable and reliable baseline of facility utilization.

The averaged data confirms a persistent bimodal usage pattern, with a pronounced morning peak at 8:00 AM (notably 181 joggers on Tuesdays) and an intense evening surge starting at 6:00 PM. Crucially, this longitudinal average

mathematically verifies that the inner lanes' Functional Capacity of 181 joggers is chronically breached during the 8:00 PM critical window across multiple days of the week, with the highest sustained density observed on Sunday (234 joggers) and Friday (226 joggers). This definitively proves that the extreme overcrowding observed in individual weeks is not a sporadic event, but a systemic operational condition of the facility.

2.1.3.1 Daily Jogger Volume Analysis

The total daily headcount was aggregated across the 21-day observation period to identify specific high-traffic days and establish a reliable longitudinal baseline. As presented in Figure 11, the facility exhibits a distinct and dynamic usage hierarchy.

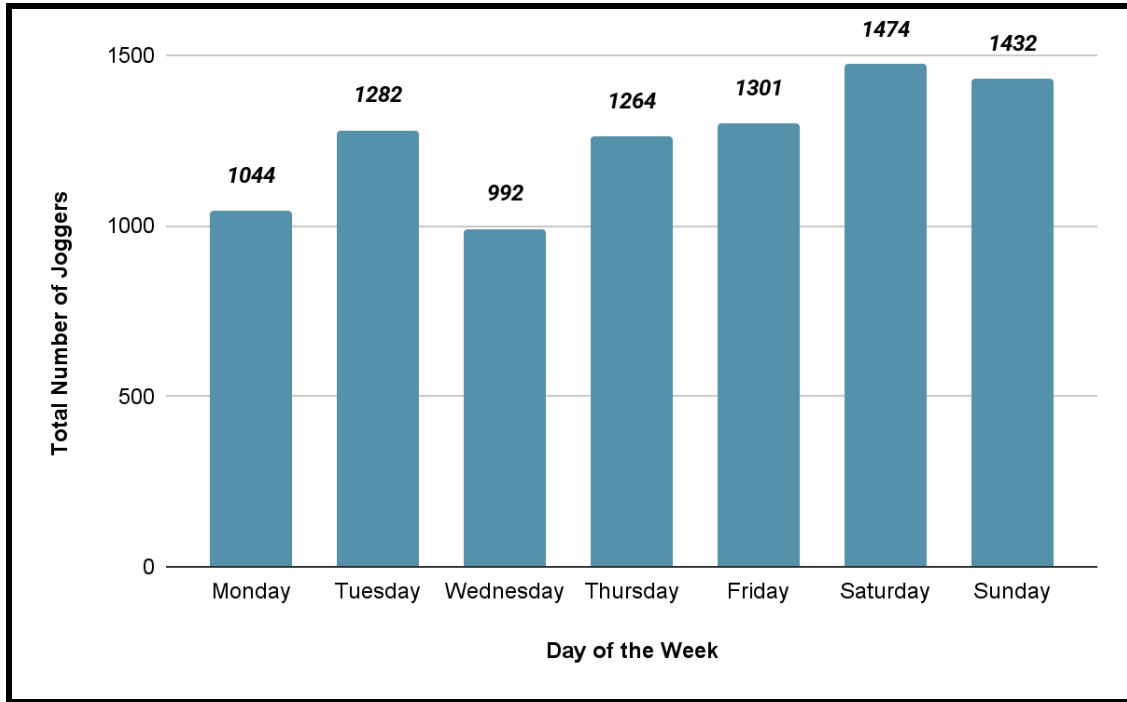


Figure 11. Three-Week Average Daily Jogger Turnout

The quantitative analysis of the three-week daily averages presents a significantly evolved operational profile compared to preliminary single-week observations. The expanded data reveals that while Tuesday (averaging 1,282 joggers) and Thursday (1,264 joggers) remain dominant during the mid-week, the absolute highest overall turnout occurs during the weekends. Saturday and Sunday recorded the peak average volumes of the week, with 1,474 and 1,432 joggers respectively, closely followed by a strong Friday turnout (1,301 joggers). This structural shift proves the necessity of the longitudinal approach, as it successfully smoothed out early-week anomalies and captured the facility's true capacity strain.

This distribution demonstrates that the CCSC Oval effectively serves a dual functional purpose. From Monday through Thursday, the facility acts as a utilitarian,

regimen-based fitness hub where usage is heavily dictated by the constraints of standard work and academic schedules. Conversely, from Friday evening through Sunday, it transitions into a high-volume public recreational space, accommodating a broader demographic with increased leisure time. Consequently, operational resources, such as track marshals and ticket booth personnel, must be strategically scaled to handle not only the compressed post-work peaks of the mid-week but also the sustained, higher-volume influx characteristic of the weekends.

To visualize the specific time-based congestion and flow dynamics underlying these daily totals, Figure 12 compares the aggregated average hourly turnout between Weekdays (Monday–Friday) and Weekends (Saturday–Sunday). This categorical aggregation strategy was adopted to mathematically isolate and compare the distinct behavioral patterns of the routine workweek crowds against the high-volume weekend users, deriving meaningful operational flow patterns that inform targeted scheduling policies.

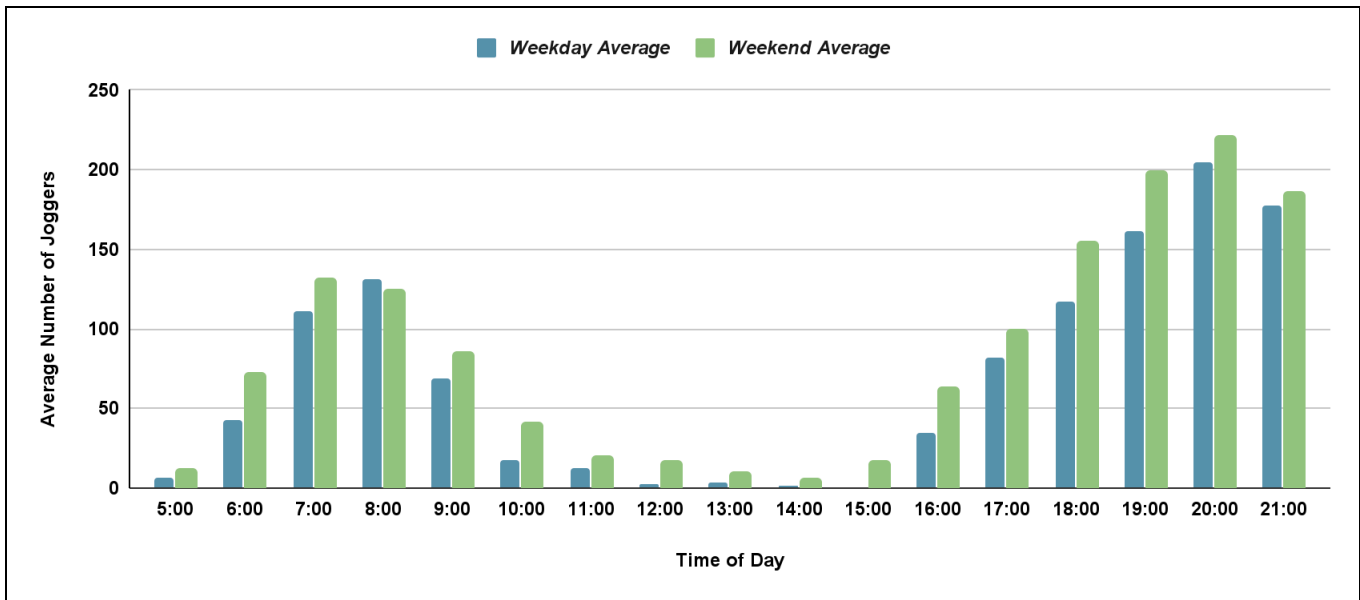


Figure 12. Three-Week Average Hourly Jogger Utilization (Weekday vs. Weekend)

The grouped bar chart provides a granular view of the facility's temporal usage over the three-week observation period, highlighting a substantial disparity in jogger density depending on the time of day. The most prominent feature across both weekday and weekend data is the sharp exponential increase in volume beginning at 5:00 PM. This evening influx escalates continuously until it reaches a daily maximum at 8:00 PM. During this critical window, the track accommodates a weekday average of 204.4 joggers and an even higher weekend average of 220.5 joggers.

When viewed through the framework of Pedestrian Level of Service (LOS) and the facility's functional inner-lane capacity of 181 joggers, this 8:00 PM peak marks a severe operational failure. At this specific density, the facility operates at LOS E–F ("Forced Flow"). The physical space available per user decreases to a point where the 2.2-meter safety cushion is completely breached, requiring joggers to drastically reduce

their speed or execute unsafe lane changes to navigate around others. This congestion presents a tangible constraint on the types of workouts that can be safely or effectively performed during these evening hours. Furthermore, the longitudinal data reveals a critical shift from initial observations: weekend evenings (averaging 220.5 joggers at 8:00 PM) now mathematically outpace weekday evenings in terms of pure density, demanding an immediate re-allocation of weekend marshal deployment.

In addition to the evening surge, the data confirms a distinct bimodal usage pattern, evidenced by a secondary peak occurring between 6:00 AM and 8:00 AM. During this morning window, the facility consistently records robust turnouts, peaking at an average of 131.6 joggers on weekdays (at 8:00 AM) and 131.0 joggers on weekends (at 7:00 AM). While this morning volume remains within the theoretical Stable Flow limits (LOS C) and does not reach the critical saturation levels of the evening rush, it demonstrates that the facility faces capacity challenges at two distinct intervals during the operational day.

Conversely, the period between 10:00 AM and 3:00 PM represents a profound Dead Zone. During this five-hour midday block, utilization drops precipitously, frequently resulting in single-digit user counts (e.g., a weekday average of merely 0.8 joggers at 3:00 PM). This drastic fluctuation, from near-zero activity at midday to massive, capacity-breaking density in the evening, indicates a significant inefficiency in the temporal distribution of facility usage. It confirms that the operational strain on the

CCSC Oval is not caused by an overall lack of facility space, but by the extreme temporal synchronization of its users.

2.1.4 Staff-Reported Peak Operational Hours

To supplement the quantitative headcount data with qualitative operational insights, the study utilized a purposive sampling method to select key facility personnel. The respondents were categorized into two primary groups: the Track Marshals, who maintain a continuous physical presence on the active ground, and the Executive Director, who provides strategic oversight across the entire facility. Unlike administrative personnel removed from daily operations, these individuals possess the direct observational experience necessary to assess crowd dynamics and safety incidents.

Table 9. Profile of Operational Staff Respondents

Operational Role / Designation	Primary Station/Zone	Number of Respondents (n)
Executive Director	Office / Roving	1
Track Marshals	Oval Track / Infield	4
Total Respondents		6

As detailed in Table 9, the respondent pool was strategically concentrated to capture perspectives from the primary point of activity. The sample consists primarily of Track Marshals, ensuring that data was gathered from personnel responsible for the direct, minute-by-minute management of the oval track. Additionally, the inclusion of the Executive Director provides a comprehensive perspective. While the Director's duties require movement between the administrative office and various external facility zones,

his roving supervision allows for the corroboration of the tactical, on-the-ground observations recorded by the marshals.

A critical refinement in this methodology is the role of the Executive Director as the Validation Authority. Following the conclusion of the 14-day data collection period, the consolidated headcount results were presented to the Director for authentication. This process ensures that the gathered data aligns with the established institutional knowledge and historical operational patterns observed from a management level.

To obtain a granular understanding of facility congestion, staff respondents were asked to log specific time windows that constitute Peak Operational Hours, defined as intervals requiring maximum vigilance and active crowd management.

Table 10. Individual Staff Reports on Peak Operational Hours

Respondent ID	Operational Role	Station Location	/	Reported Interval	Peak
R1	Executive Director	Admin Office		5:00 PM – 6:00 PM	
R2	Track Marshal	Oval Track		5:00 PM – 6:00 PM	
R3	Track Marshal	Oval Track		5:00 PM – 7:00 PM	
R4	Track Marshal	Oval Track		6:00 PM – 7:00 PM	
R5	Track Marshal	Oval Track		5:00 PM – 7:00 PM	

The individual staff reports in Table 10 illustrate a unified consensus regarding the evening Peak Density Window. By focusing on personnel with direct track-level

responsibility and management-level oversight, the data demonstrates a clear correlation between user influx and track saturation. The majority of respondents identify the 5:00 PM to 7:00 PM interval as the critical period for facility load.

Furthermore, the data shows a distinct pattern of time-based development throughout the entire evening period. The facility access data shows that students and early commuters start entering the building at 3:00 PM because this time marks the beginning of the "Ingress Phase" period. The Track Marshals (R2 R3 R5) provide their first report of operational performance results which demonstrate that actual capacity starts at 5:00 PM because it takes time for people to enter the facility until tracks reach capacity.

The Executive Director (R1) and Respondent R2 report that their peak period lasts for one hour between 5:00 PM and 6:00 PM while other Marshals believe that traffic will continue until 7:00 PM. The density reaches "crisis" levels during the first hour but requires ground crew members to handle operational tasks which last longer than the time administrators need to control the crowd and track discipline. The existing system needs improvement because it requires automated solutions which track active data throughout all peak hours instead of depending on hourly peak times.

2.1.5 Self-Reported Usage of Joggers

To contextualize the self-reported data, the survey first established the habitual nature of the respondents' facility usage. Understanding how often these individuals utilize the CCSC Oval is essential for determining whether the observed peak hours are driven by consistent, high-frequency users or sporadic visitors.

Table 11. Distribution of Respondents by Jogging Frequency

Frequency of Visits (Per Week)	Number of Respondents (n)	Percentage (%)
7	16	13.79%
6	8	6.9%
5	11	9.48%
4	9	7.76%
3	12	10.34%
2	32	27.59%
1	28	24.14%
Total	116	100%

The profile of the expanded respondent pool (n = 116), as shown in Table 11, indicates a usage pattern heavily weighted toward regular, non-daily intervals. The largest segment of the population (34.5%) visits the facility exactly 2 times per week, while another significant cohort (19.0%) visits 3 times per week. This frequency distribution provides a clear behavioral explanation for the routine spikes observed in the manual headcounts. Unlike daily joggers who distribute their load evenly across the week, "2-day" and "3-day" users tend to cluster their visits on specific alternating

schedules, driving predictable density spikes on recurring days. Furthermore, the data reveals that nearly 65.6% of the respondents visit the track 2 to 4 times a week, compared to only 25.9% who visit once. This dominance of habitual users confirms that the CCSC Oval functions strictly as a routine-based fitness hub. The congestion issues, therefore, are not caused by unpredictable random crowds, but by the synchronized schedules of a dedicated user base whose routines overlap consistently. To identify the relationship between the users' ideal schedules and their actual usage behaviors, respondents were asked to indicate their preferred time and day for jogging.

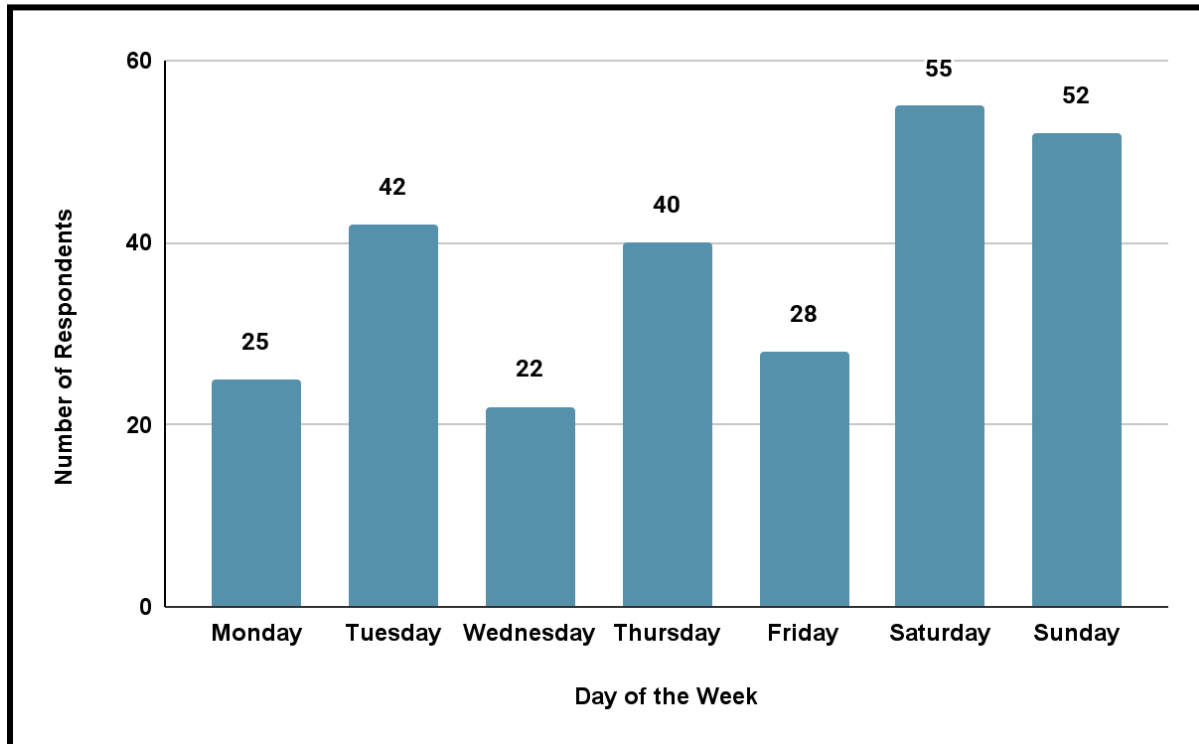


Figure 13. Self-Reported Preferred Jogging Days

The analysis of preferred jogging days presented in Figure 13 exposes a direct alignment between user intent and the established operational reality derived from the longitudinal headcounts. The expanded three-week manual observations definitively

identified the weekend as the facility's highest-volume period, which is perfectly mirrored by the self-reported survey data. Respondents overwhelmingly identified Saturday (n = 55) and Sunday (n = 52) as their most desired days for utilization. This synchronization between preference and actual turnout indicates that when users are free from the constraints of their professional or academic schedules, they default to weekend usage. Consequently, the high volume of runners observed on the weekends is a reflection of genuine leisure preference. Conversely, the secondary peaks observed on Tuesday (n = 42) and Thursday (n = 40) represent a scheduling necessity for the habitual "2-day" and "3-day" users who must fit routine exercise into their standard workweek. This finding is critical for facility management, as it indicates that the Oval manages two entirely different demographic intents: it functions as a utilitarian fitness hub during the mid-week and transitions into a highly sought-after recreational destination on the weekends, necessitating distinct crowd-control policies for each phase.

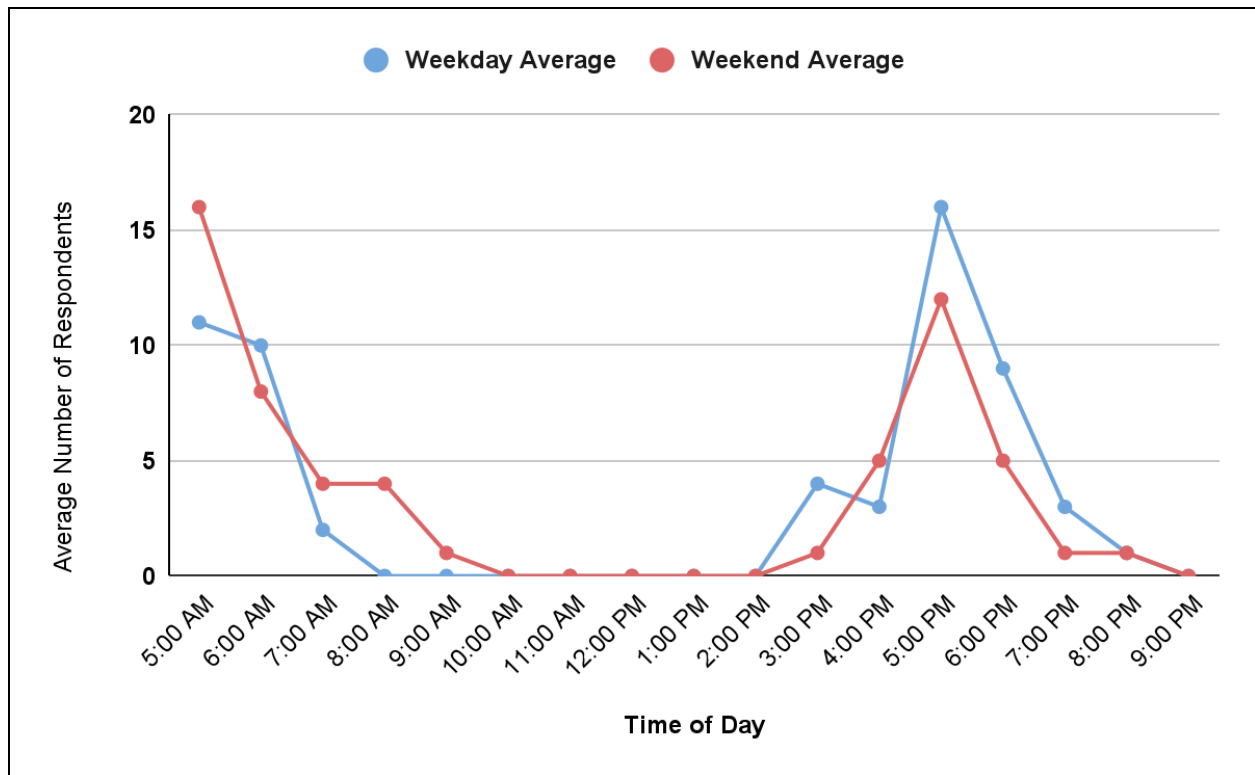


Figure 14. Self-Reported Preferred Jogging Hours

In terms of temporal distribution, Figure 14 illustrates a high degree of synchronization in user preferences that perfectly maps to the facility's observed bimodal usage patterns. Driven by the stratified sampling design, the data captures two distinct arrival preferences: a pronounced morning intent centered at 5:00 AM, and a massive evening surge at 5:00 PM.

When comparing weekday and weekend preferences, a subtle but critical behavioral shift emerges. During the workweek, arrivals are heavily concentrated strictly at 5:00 AM and 5:00 PM, reflecting the inelastic constraints of standard 8-to-5 professional and academic schedules. However, on the weekends, user preferences exhibit a wider temporal spread. Morning weekend joggers show an increased

willingness to arrive later (at 7:00 AM and 8:00 AM), while evening weekend joggers begin arriving slightly earlier (at 4:00 PM). This indicates that when users are free from schedule constraints, their arrival times disperse, which directly correlates to the higher but more sustained track volumes observed in the weekend manual headcounts.

Most importantly, the 5:00 PM self-reported "ideal arrival time" provides vital context for understanding the operational mechanics of the facility's most severe congestion. While the objective manual headcount data recorded a physical congestion period that escalates well into the evening, peaking in maximum track density at 8:00 PM, the survey data reveals that the psychological preference for arriving drastically diminishes after 6:00 PM.

This mathematical discrepancy clarifies the anatomy of the evening congestion: the facility does not experience a continuous, rising stream of new arrivals leading up to 8:00 PM; rather, the extreme density is the result of a cumulative lag effect. The massive cohort of joggers arriving at their preferred time of 5:00 PM remains on the track for extended sessions, overlapping directly with the subsequent, smaller waves of users entering at 6:00 PM and 7:00 PM. Therefore, the critical density observed at 8:00 PM is not caused by a late-night rush of new users, but by the continuous accumulation and prolonged dwell time of the afternoon crowd. This distinction, separating the Arrival Peak from the Occupancy Peak, is fundamental for policy development. It proves that technological and operational strategies aimed at resolving the 8:00 PM track saturation

must proactively manage the flow, lane zoning, and turnover of the users who arrive hours earlier.

2.1.6 Definitive Peak Periods

To validate the study's findings, Table 8 triangulates the peak periods identified by the three independent data sources: objective manual headcounts, subjective staff observations, and self-reported jogger preferences. This comparison reveals a chronological progression in the facility's congestion, demonstrating how the peak period evolves through different stages of operation rather than occurring as a single static event.

Table 12. Triangulation of Peak Periods by Data Source

Data Source	Identified Peak Time	Operational Definition (Context)
Manual Count (Objective)	6:00 PM – 8:00 PM	The period of maximum physical congestion (Headcount Volume).
Staff Observation (Subjective)	5:00 PM – 7:00 PM	The period requiring active crowd management (Ingress & Ticketing).
Jogger Survey (Self-Reported)	5:00 PM	The preferred start time of the joggers.

The triangulated data presented in Table 12 illustrates a clear causal chain in the development of facility congestion across both the regimen-based weekdays and the high-volume weekends. The cycle initiates with the Jogger Survey data, which identifies the 5:00 PM window as the overwhelming preferred arrival time for the majority of

evening users. This period represents the users' desire to exercise aligns with the end of the standard workday or academic classes. This concentrated arrival window is the primary catalyst for the subsequent operational challenges, as it compresses the daily influx of users into a single, narrow timeframe.

This massive surge in arrivals immediately triggers the Staff-Reported Peak, heavily impacting the Ticket Booth personnel responsible for managing the ingress queues. From 5:00 PM to 7:00 PM, the operational burden is strictly focused on the facility's perimeter. Staff members must process high-volume transactions and manage the physical flow of people entering the gates. This phase is characterized by severe transaction density and queue management stress, occurring before the track itself even reaches its maximum physical saturation.

Finally, the three-week Manual Headcounts record the highest physical volume later in the evening, specifically escalating from 6:00 PM and culminating at 8:00 PM, where track density routinely breaches the 181-jogger capacity limit. This phase represents the ultimate culmination of the accumulation process. The mathematical lag between the initial 5:00 PM arrivals and the 8:00 PM peak density occurs because the massive cohort of users who arrive at 5:00 PM typically remain on the track for extended, multi-hour workouts. As these early arrivals continue their sessions, they are systematically joined by subsequent, smaller waves of joggers arriving at 6:00 PM and 7:00 PM, leading to a cumulative saturation effect (Level of Service E-F). Consequently,

the critical density observed at 8:00 PM is not the result of a sudden, late-night influx, but rather the sustained presence of the crowds that arrived hours earlier.

Therefore, while the physical peak density is statistically highest at 8:00 PM, the operational peak effectively begins at 5:00 PM. The extreme congestion at 8:00 PM is a direct downstream consequence of the 5:00 PM arrival bottleneck and the subsequent dwell time of those users. This distinction is vital for facility management and policy formulation: it indicates that any intervention aimed at reducing the dangerous 8:00 PM track density must address the root cause, the synchronized arrival and turnover rates of the 5:00 PM crowd, rather than simply attempting to restrict facility access during the late evening hours.

2.1.7 Analysis of Occupancy Rates and Verification of Congestion Thresholds

To verify if the oval's occupancy rate has exceeded its threshold, the study refers to Table 2 and Table 8 to conduct a comparative analysis between theoretical limits and observed usage. The assessment employs Equation 5:

$$O = \frac{N_a}{N_c} \times 100 \quad (5)$$

as established in the research methodology 3, to quantify the ratio between the actual number of joggers (N_a) and the calculated capacity (N_c). This formula was selected because it provides a standardized metric for identifying congestion, where a rate exceeding 100% explicitly indicates overcrowding and reduced flow efficiency. While the facility possesses a total theoretical capacity of approximately 1,600 joggers, the inner

lanes (Lanes 1-2), which are the primary preference for runners, are limited to a functional capacity of approximately 181 joggers.

To address the temporal extent of the congestion and provide a comprehensive view of the facility's daily load, the data was analyzed using two distinct metrics: the Average Hourly Occupancy (the mean jogger load distributed across the entire 17-hour operational day) and the Peak Hourly Occupancy (the maximum sustained load during the critical 8:00 PM window). Table 13 summarizes these findings, mapping the calculated percentages to the established Pedestrian Level of Service (LOS) framework to ascertain whether the congestion is an isolated anomaly or a systemic operational failure.

Table 13. Summary of Peak Occupancy Rates by Hourly Average

Day	Average Hourly Headcount	Average Inner-Lane Occupancy	Average LOS (Track State)
Monday	61	33.7%	LOS A-B
Tuesday	75	41.4%	LOS A-B
Wednesday	58	32.0%	LOS A-B
Thursday	74	40.9%	LOS A-B

Friday	77	42.5%	LOS A-B
Saturday	87	48.1%	LOS A-B
Sunday	84	46.4%	LOS A-B

The data in Table 13 reveals a significant statistical divergence that highlights the danger of relying solely on daily averages for facility management. When evaluating the Average Hourly Occupancy, the inner lanes appear to operate well within safe limits, ranging from 32.0% to 48.1%. Under the established pedestrian framework, this corresponds to LOS A-B (Free Flow), suggesting ample space, unrestricted pacing, and no overtaking conflicts. However, this daily average is a mathematical artifact caused by the facility's massive five-hour midday dead-zone (10:00 AM to 3:00 PM). The near-zero utilization during these daytime hours artificially dilutes the data, masking the severe congestion that occurs during the actual operational peaks.

Table 14. Summary of Occupancy Rates by Morning Peak Average (5AM - 8AM)

Day	Average Hourly Headcount	Average Inner-Lane Occupancy	Average LOS (Track State)	Average Facility Occupancy (Total)
Monday	61	33.7%	LOS A-B	3.8%
Tuesday	75	41.4%	LOS A-B	4.7%
Wednesday	58	32.0%	LOS A-B	3.6%
Thursday	74	40.9%	LOS A-B	4.6%
Friday	77	42.5%	LOS A-B	4.8%
Saturday	87	48.1%	LOS A-B	5.4%
Sunday	84	46.4%	LOS A-B	5.3%

Table 14 evaluates the facility's sustained operational load during the four-hour morning window (5:00 AM to 8:00 AM). The block averages indicate that the morning peak, while representing a distinct bimodal surge, maintains a highly manageable density across all seven days. The highest sustained morning loads occur on Fridays

and Saturdays, averaging 90 joggers over the four-hour period, which translates to a 49.7% Inner-Lane Occupancy.

Under the established pedestrian framework, this continuous morning block operates reliably at LOS A-B (Free Flow). Because the arrivals are spread comfortably across the four hours, users experience ample space, unrestricted pacing, and minimal overtaking conflicts. This data mathematically confirms that the facility's morning operations—even on peak days like Tuesday and Thursday, do not suffer from the severe, cumulative saturation that plagues the evening shift, and therefore do not require the same aggressive crowd-control interventions.

Table 15. Summary of Occupancy Rates by Evening Peak Average (5:00 PM – 8:00 PM)

Day	Evening Average Headcount	Block	Average Inner-Lane Occupancy	Average LOS (Track State)	Average Facility Occupancy (Total)
Monday	131		72.4%	LOS C	8.2%
Tuesday	151		83.4%	LOS C-D	9.4%
Wednesday	133		73.5%	LOS C	8.3%
Thursday	148		81.8%	LOS C-D	9.2%

Friday	145	80.1%	LOS C-D	9.1%
Saturday	172	95.0%	LOS D-E	10.7%
Sunday	164	90.6%	LOS D-E	10.2%

Table 15 evaluates the extreme, sustained saturation experienced during the four-hour evening influx (5:00 PM to 8:00 PM). By calculating the block average, the data exposes a profound shift in facility dynamics compared to the morning hours. Throughout the standard workweek (Monday–Friday), the track sustains an average four-hour load that continuously hovers around 80% capacity (LOS C-D). This indicates a prolonged Restricted Flow state where the sheer volume of users consistently impedes optimal running paces for hours at a time.

However, the most critical operational failure occurs during the weekend evening blocks. On Saturdays and Sundays, the track sustains an exhausting four-hour average of 95.0% and 90.6% inner-lane occupancy, respectively (LOS D-E). This means that for four consecutive hours, the facility is pushed to the absolute brink of its physical breaking point. It is vital to note that while this four-hour *average* sits just below the 100% threshold, it acts as the baseline pressure that inevitably pushes the specific 8:00 PM maximum hour into severe forced flow (>120% capacity).

Furthermore, the Average Facility Occupancy metric proves that even during these grueling four-hour evening blocks, the overall stadium barely utilizes 10% of its spatial footprint. This definitively validates that the evening crisis is a highly localized failure of track lane capacity caused by sustained, overlapping dwell times, necessitating immediate technical monitoring and physical lane enforcement.

Table 16. Comparison of Total Daily Volume to Maximum Facility Capacity

Day	Total Volume (Joggers)	Daily Volume vs. Max Capacity	Theoretical Daily LOS (Facility State)
Monday	1,044	65.3%	LOS C
Tuesday	1,282	80.1%	LOS D
Wednesday	992	62.0%	LOS C
Thursday	1,264	79.0%	LOS D
Friday	1,301	81.3%	LOS D
Saturday	1,474	92.1%	LOS E
Sunday	1,432	89.5%	LOS D-E

Table 16 compares the macro-level Total Daily Volume against the facility's absolute maximum spatial capacity of 1,600 concurrent users. This comparative analysis exposes a profound architectural and operational paradox: the CCSC Oval is a massive venue that is fundamentally bottlenecked by a highly specific use-case.

The data mathematically demonstrates that the total daily throughput of the facility never exceeds its instantaneous holding limit. Even on Saturday, the absolute busiest day of the week, logging an immense 1,474 individual joggers, the entire day's volume represents 92.1% of the stadium's total capacity (Theoretical LOS E). This means that theoretically, the facility possesses enough physical space to accommodate every single jogger who visits on a Saturday simultaneously without breaching the stadium's overarching 100% threshold.

This comparison solidifies the core finding of the operational assessment: the CCSC Oval does not suffer from a macro-level infrastructure deficit. The venue is more than large enough for the community it serves. The critical, localized track failures (LOS E-F) identified during the 8:00 PM evening peak occur exclusively because the vast majority of these users synchronize their schedules and aggressively compete for the same 181 functional slots in the inner running lanes.

Therefore, any macro-level management intervention, such as facility expansion or capping total daily ticket sales, would be a misallocation of resources. The data strictly dictates that operational policies must be surgical, utilizing real-time monitoring to enforce lane discipline, distribute peak-hour arrivals, and maximize the vastly underutilized spatial footprint of the venue's outer lanes.

2.1.4 Impact of Density on Jogger Pace and Flow Dynamics

To quantify this phenomenon at the CCSC Oval, a time-space study was extracted from the existing three-week video footage. To ensure statistical reliability and accurately represent the longitudinal data, the researchers utilized a stratified random sampling method, tracking exactly 30 joggers per shift ($n = 6$) per day, yielding a robust total of $n = 420$ tracked full laps across the synthesized seven-day baseline).

During each designated period, Morning Peak (5:00 AM – 8:00 AM) and the Evening Peak (5:00 PM – 8:00 PM), the observers isolated and timed the full-lap completion of, 10 random joggers strictly utilizing the Inner Lanes, 10 random joggers utilizing the Middle Lanes, and 10 random joggers utilizing the Outer Lanes. The average velocity v for each tracked user was calculated using the fundamental kinematic equation:

$$v = \frac{d}{t} \quad (6)$$

where d represents the adjusted staggered distance of the specific lane utilized (ranging from 400 meters for Lane 1 up to approximately 450 meters for Lane 8), and t represents the full lap completion time in seconds. These velocities were then converted into the standard athletic metric of minutes per kilometer (min/km).

Table 17. Average Full-Lap Pace by Lane Position (Morning vs. Evening Peak)

Day	Shift	Inner Lanes (Lanes 1-2)	Middle Lanes (Lanes 3-5)	Outer Lanes (Lanes 6-8)
Monday	Morning	5:43 min/km	7:17 min/km	11:21 min/km
	Evening	7:14 min/km	8:33 min/km	12:01 min/km
Tuesday	Morning	5:42 min/km	7:18 min/km	11:40 min/km
	Evening	8:11 min/km	9:29 min/km	12:22 min/km
Wednesday	Morning	5:23 min/km	7:14 min/km	11:12 min/km
	Evening	7:12 min/km	8:44 min/km	12:10 min/km
Thursday	Morning	5:53 min/km	7:33 min/km	11:42 min/km
	Evening	8:22 min/km	9:12 min/km	12:13 min/km
Friday	Morning	6:01 min/km	7:28 min/km	11:39 min/km
	Evening	9:36 min/km	9:44 min/km	11:40 min/km
Saturday	Morning	6:10 min/km	7:18 min/km	12:11 min/km
	Evening	7:12 min/km	9:40 min/km	12:50 min/km
Sunday	Morning	6:05 min/km	7:23 min/km	12:12 min/km
	Evening	8:34 min/km	10:02 min/km	13:11 min/km

The kinematic data in Table 17 provides empirical evidence of severe functional congestion, revealing that the track's spatial saturation does not affect all users equally.

During the moderate morning hours, clear lane delineation is naturally maintained. Fast runners in the Inner Lanes successfully sustain an unimpeded cardiovascular pace (averaging under 6:00 min/km), while the Middle and Outer lanes scale appropriately for slower joggers and walkers.

However, during the massive evening influx, a cascading mechanical failure of track flow occurs. The most profound degradation is observed in the Inner Lanes. For example, on Sunday evenings, the facility's highest volume period, the Inner Lane pace drops catastrophically from 6:05 min/km in the morning to 8:34 min/km in the evening. Observational review confirms this deceleration is entirely forced by spatial limits. The sheer density of users occupying Lanes 1 and 2 eliminates safe trailing distances, completely neutralizing the intended purpose of the "fast lanes."

Furthermore, as the Inner Lanes reach critical mass, faster joggers attempt to execute continuous overtaking maneuvers by shifting outward. This lateral displacement artificially congests the Middle Lanes, causing their average pace to degrade correspondingly. Conversely, the Outer Lanes experience the least severe pace degradation (adding roughly 1 minute per kilometer), as the users in these lanes are already traveling at a walking velocity and are less impacted by forward physical obstruction.

This multi-lane analysis proves that the evening congestion is not merely a quantitative headcount issue, but a qualitative failure of facility zoning. The extreme

density forces all lanes to homogenize into a uniform, forced-flow walking pace, definitively validating the need for strict, automated capacity management and active lane segregation policies.

2.2 Assessing Operational Challenges in Current People-Counting Practice

This section addresses the second statement of the problem, focusing on the operational challenges faced by CCSC personnel in implementing people-counting and crowd control procedures. Data was derived from semi-structured interviews and questionnaires with staff members responsible for monitoring, documentation, and safety enforcement. Through thematic analysis, the responses were categorized to identify systemic limitations in the current manual counting method.

2.2.1 Selection and Demography of Respondents

Respondents were selected through purposive sampling, specifically targeting the Track Marshals whose primary job description involves the direct, on-ground monitoring of the oval track. As detailed in Table 6, this focused group represents the sole on-track gatekeepers of the facility, providing the only existing record of how crowd-related incidents and peak-hour surges are managed in real-time. By limiting the survey to these specific personnel, the researchers ensured that the qualitative data reflects the actual conditions experienced on the running lanes.

2.2.2 Quantitative Presentation and Mathematical Treatment

To evaluate the severity of operational challenges, the survey utilized a 5-point Likert Scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The researchers calculated the Weighted Mean to determine the average staff sentiment and the Standard Deviation to measure the consistency of their responses. For the critical indicator of Counting Accuracy during Peak Hours, the mathematical derivation is as follows:

Weighted Mean Calculation:

$$\text{Sum} = (3 \times 1) + (3 \times 2) + (2 \times 3) = 15$$

$$\bar{x} = \frac{\sum(f \cdot w)}{N} = \frac{15}{8} = 1.88$$

Standard Deviation Calculation:

$$\text{Variance } (s^2) = \frac{\sum(x - \bar{x})^2}{N - 1}$$

$$\text{Variance } (s^2) = \frac{3(1 - 1.88)^2 + 3(2 - 1.88)^2 + 2(3 - 1.88)^2}{8 - 1}$$

$$\text{Variance } (s^2) = \frac{3(0.7744) + 3(0.0144) + 2(1.2544)}{7} = \frac{4.84}{7} = 0.6914$$

$$s = \sqrt{0.6914} = 0.83$$

Table 18. Assessment of Operational Challenges by CCSC Personnel

Indicator	Weighted Mean ($\bar{x}$)	SD (s)	Verbal Interpretation
1. Sufficiency of manual counting tools	2.13	0.83	Disagree
2. Accuracy of counting during peak hours	1.88	0.83	Strongly Disagree
3. Reliability of communication tools	3.50	1.07	Neutral / Agree
4. Ease of data retrieval and reconciliation	1.88	0.64	Strongly Disagree
Composite Mean	2.35		Disagree

2.2.3 Operational Interpretation of Results

The quantitative data indicates a systemic failure in the manual counting protocol during high-traffic periods. The lowest mean scores (1.88) regarding Accuracy and Data Retrieval confirm that the staff feels the current method is ineffective when the facility is most vulnerable. While communication tools such as radios received a neutral-to-high score, the high Standard Deviation (1.07) suggests a lack of uniformity in their use; some staff members rely on radios while others continue to use verbal shouting. Within the context of this study, congestion is operationally defined as the state where the ingress queue at the Front Gate exceeds 15 persons, causing a physical bottleneck. Peak Hours refers specifically to the 5:00 PM to 8:00 PM window where the influx

exceeds manual tallying capacity. Finally, Overcrowding is defined as the point where the inner lanes of the oval are occupied by more than two joggers abreast, effectively obstructing the line of sight for Track Marshals and making visual counting physically impossible.

2.2.4 Narrative Compilation of Operational Findings

The qualitative data from open-ended survey responses reveals that the visual observation method is primarily reactive. Personnel reported that they often rely on end-of-shift summaries based on ticket stubs, which results in a lack of real-time occupancy data during peak activity. To provide a comprehensive view of these systemic failures, the researchers categorized the staff's first-hand experiences into three core operational themes.

Table 19. Thematic Analysis of Manual Counting Constraints

Theme Category	Operational Context	Key Observations (Verbatim)	Staff English Translation / Summary
Operational & Manual Limitations	Inherent fallibility of visual tracking and human error.	• "Manual ra man, mao nga masayop mi usahay sa ihap." (R3) • "Dali ra kaayo moagi ang mga joggers, di nako matanaw tanan." (R5) • "Mag basi rami sa among pag bana-bana." (R2)	• "It is just manual, that is why we sometimes make mistakes in the count." • "The joggers pass by very quickly; I cannot see everyone." • "We just base it on our own estimations."
Data Blind Spots	Exclusion of specific user groups from official logs.	• "Dili namo ma apil ug ihap ang mga player ug estudyante sa Abellana." (R3)	• "Students and athletes are not recorded in the logs." • "We are unable to include the players and students from Abellana in the count."
Systemic Inefficiency	Bottlenecks caused by slow processing and lane non-compliance.	• "Hinay og proceso... dili tanan amo ma ticketan." (R4) • "Usahay ang mga mag dagan di kabalo sa mga balaod sa lanes." (R5)	• "The process is slow... we cannot issue tickets to everyone." • "Sometimes the runners do not know the rules regarding the lanes."

The data reveals that the primary challenge facing CCSC staff is the inherent fallibility of the manual counting system. The Track Marshals identified manual counting as a primary source of error, noting that the sheer volume of entrants during peak hours particularly the evening session forces them to rely on estimation (pag bana-bana)

rather than precision. This is especially evident during periods where the density exceeds the human capacity to visually record each individual simultaneously.

Furthermore, a critical Data Blind Spot was confirmed regarding non-paying users. As noted by respondent R3, athletes and students often bypass the ticketing process. Because the facility's official attendance records are based solely on ticket sales, the current practice systematically underreports actual crowd density. This discrepancy proves that the current manual method is unable to provide the ground truth of facility usage, highlighting the need for an automated tracking solution that monitors physical presence rather than transactional logs.

2.3 Jogger Perception of Crowd Management

The camera system measured jogger numbers to produce accurate data, but user experience needs additional research beyond numerical measurement. The research needed to collect qualitative information because high user volume does not always lead to customer discontent, which required funding to establish primary user needs. The section presents detailed information about respondent comments by separating them into two separate time frames which include the Morning Session that runs from 5:00 AM to 8:00 AM and the Evening Session that runs from 5:00 PM to 8:00 PM. The researchers use this segmentation method to test whether the camera analysis achieves its maximum points which results in different perceptions of overcrowding and safety hazards and performance limitations throughout the entire day. The researchers

used a survey questionnaire which they distributed at the CCSC Oval to measure how users viewed their experience.

The instrument utilized a 5-Point Likert Scale, where respondents rated their level of agreement with statements regarding crowd density, space availability, and safety. The scale ranged from 1 (Strongly Disagree) to 5 (Strongly Agree). The study included open-ended questions to gather public attitudes about crowd numbers. The study selected jogger participants through specific criteria which required them to hold a current Jogger's Pass and use the oval track for running activities. The researchers used Convenience Sampling by approaching joggers who were present on-site at their actual visit times. The study included 116 joggers who participated in the study, which included both morning and evening groups.

To interpret the survey responses, the data was analyzed using the statistical tools defined in the methodology. The Weighted Mean ($\bar{X}$) was calculated to determine the central tendency of respondent sentiment, following the formula in Equation 2. Additionally, the Standard Deviation (SD) was computed using Equation 3 to measure the variability of responses and determine if the joggers' opinions were consistent or diverse.

$$\bar{X} = \frac{\sum X}{n} \quad (2)$$

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}} \quad (3)$$

The following section presents the analysis for each indicator based on these calculations.

Table 20. Perception of Crowd Density

Item 1	Session	Mean Score	SD	Interpretation
I find the crowd density on the oval appropriate during my usual jogging times.	Morning	3.48	0.92	Neutral
	Evening	2.58	1.03	Disagree / Dissatisfied

The morning respondents yielded a mean score of 3.48 with a standard deviation of 0.92, falling within the Neutral range. This indicates that while morning joggers do not perceive the crowd as overwhelming, they also do not find the density completely ideal, suggesting that even early-hour volumes are beginning to approach the limits of user comfort. In contrast, the evening respondents yielded a mean score of 2.58, falling within the Disagree range.

This divergence serves as critical user-validation of the quantitative camera data. It confirms that the high volume of people captured during the 5:00 PM to 8:00 PM window is not merely a statistical figure, but a tangible felt experience of overcrowding

that users consider inappropriate for a sports facility. The standard deviation of 1.03 in the evening suggests a wider variance in user tolerance, yet the low mean score overall reinforces the necessity of an automated system to monitor these surges. While the morning session remains tolerable, the evening data confirms that the facility's current manual practices fail to maintain a density that users perceive as appropriate for athletic activities.

Table 21. Space Availability

<i>Item 2</i>	<i>Session</i>	<i>Mean Score</i>	<i>SD</i>	<i>Interpretation</i>
<i>There is enough space on the oval for me to jog comfortably without feeling crowded.</i>	Morning	3.38	0.95	<i>Neutral</i>
	Evening	2.73	1.15	<i>Disagree / Dissatisfied</i>

The morning respondents yielded a mean score of 3.38 with a standard deviation of 0.95, falling within the Neutral range. While this is significantly higher than the evening score, it indicates that even during lower-volume morning hours, joggers do not explicitly agree that the space is sufficient; instead, they find it merely tolerable. In contrast, the evening respondents yielded a mean score of 2.73, explicitly disagreeing that there is enough space to jog comfortably.

This indicator speaks directly to the physical capacity of the lanes. Unlike a casual park where crowds might be tolerable, a track oval requires specific spatial dimensions approximately 2 meters of longitudinal space per person to allow for safe, rhythmic movement. The neutral rating in the morning suggests that while the personal bubble is less frequently violated than in the evening, users still perceive the presence of others as a constraint on their movement. The negative rating in the evening confirms a critical failure in occupancy management, where the lack of space forces runners to break their stride, slow down, or maneuver abruptly to avoid collisions. This data proves that the facility allows entry beyond what the physical dimensions of the track can comfortably support, creating a persistent bottleneck effect that ranges from a neutral inconvenience in the morning to a significant disruption in the evening.

Table 22. Consistent Performance

Item 3	Session	Mean Score	SD	Interpretation
The condition of the oval supports my ability to maintain consistent jogging performance.	Morning	3.32	0.98	Neutral
	Evening	2.55	1.23	Disagree / Dissatisfied

The morning respondents yielded a mean score of 3.32 with a standard deviation of 0.98, falling within the Neutral range. This indicates that while morning joggers are less obstructed than evening joggers, they still do not fully agree that the environment is

optimal for consistent athletic performance, suggesting that even moderate morning volumes present minor rhythmic interruptions. In contrast, the evening respondents yielded a mean score of 2.55, explicitly disagreeing that the current conditions support consistent jogging performance.

This finding links the crowd conditions directly to the core functionality of the facility. The CCSC Oval is an athletic venue designed for continuous physical activity; however, the respondents indicate that the current conditions particularly in the evening prevent them from maintaining a steady pace. For the active users identified in the study, the inability to maintain a consistent cadence due to human traffic is a major source of frustration. This effectively transforms a structured workout into an obstacle course, negating the primary purpose of the track. The disparity between the neutral morning score and the negative evening score confirms that as density increases, the functional utility of the track for serious training significantly degrades.

Table 23. Track Etiquette and Discipline

Item 4		Session	Mean Score	SD	Interpretation
Other joggers generally follow proper track etiquette, such as lane discipline and passing rules.		Morning	3.02	1.05	Neutral
		Evening	2.30	1.10	Disagree / Dissatisfied

The morning respondents yielded a mean score of 3.02, while the evening respondents yielded the lowest score in the entire survey at 2.30. This signals a strong disagreement across both groups regarding user discipline, though the dissatisfaction is significantly more acute during the evening rush.

This result adds a crucial layer to the operational problem: the issue is not merely the quantity of people on the track, but their spatial behavior and adherence to established norms. Respondents report frequent violations of track etiquette, such as slower users walking in the inner fast lanes or groups stopping abruptly, which forces active joggers to break their stride. The low score in the evening serves as an indictment of current crowd control measures, implying that the manual monitoring system is largely passive. While staff may count entrants at the gate, there is a clear deficiency in active marshaling on the track itself to enforce lane discipline. This breakdown in etiquette exacerbates the feeling of congestion and safety risks, confirming that manual oversight is insufficient to regulate user behavior when density reaches peak levels.

Table 24. Communication of Rules

Item 5	Session	Mean Score	SD	Interpretation
The rules and guidelines for using the oval are clearly communicated to me.	Morning	3.30	1.25	Neutral
	Evening	3.24	1.39	Neutral

Both the morning and evening respondents yielded scores within the Neutral range, at 3.30 and 3.24 respectively. These are the highest mean scores among all perception indicators, yet they still fall short of a "Satisfied" or "Agree" rating. This neutrality suggests a significant gap between the passive existence of rules and the active communication of those rules to the public.

While users acknowledge that rules and guidelines exist through static signage at the entrance, the lack of a higher score indicates that the communication is not impactful enough to influence user behavior. The consistency of these neutral scores across both sessions reveals a systemic issue: the current manual system focuses on rule-making rather than rule-reinforcement. The significant gap between this Neutral score and the very low score for Track Etiquette (Item 4) proves that users may be aware of the rules, but without active, data-driven communication or visible management on the track, compliance remains stagnant. This supports the need for an automated system that can trigger real-time alerts and active marshaling when density thresholds are reached.

Table 25. Overall Environment

Item 6	Session	Mean Score	SD	Interpretation
The overall jogging environment on the CCSC oval meets my expectations.	Morning	3.42	0.95	Neutral
	Evening	2.85	1.12	Disagree / Dissatisfied

The cumulative impact of the previous factors results in an overall environment rating of 2.85 (Disagree). This score is the summary judgment of the respondents' experience. It reflects that the combination of physical overcrowding, lack of personal space, and poor user discipline creates an unsatisfactory environment. For a group of regular pass holders who pay for access, this dissatisfaction is significant. It underscores that the current manual headcount practice is insufficient because it treats crowd management as a gatekeeping task (counting tickets) rather than a holistic operational task (ensuring a safe, high-quality running environment).

Table 26. Summary of Jogger Perception on Crowd Management

Indicator Statement	Morning ($\bar{x}$)	Evening ($\bar{x}$)	Total Composite Mean	Verbal Interpretation
A. Crowd Density and Space	3.43	2.62	3.01	Neutral
1. Appropriate Density	3.48	2.58	2.03	Neutral
2. Enough Space to Jog	3.38	2.73	3.06	Neutral
3. Supports Performance	3.42	2.55	2.99	Disagree
B. Crowd Management of Joggers	3.16	2.77	2.96	Disagree
4. Track Etiquette Followed	3.02	2.30	2.66	Disagree
5. Rules Communicated	3.30	3.24	3.27	Neutral
C. Overall Experience	3.42	2.85	3.13	
6. Environment Expectations	3.42	2.85	3.14	Disagree
Composite Mean	3.34	2.75	3.03	Disagree

The summary table reveals a significant disparity in user perception regarding the current crowd conditions at the CCSC Oval, depending on the time of visit. While the Total Composite Mean (3.05) suggests a neutral overall sentiment, this figure is a mathematical average that masks the critical dissatisfaction during peak periods. When segmented, the Evening Composite Mean of 2.75 confirms that respondents generally disagree that the facility effectively manages crowd levels during peak hours.

Conversely, the Morning Composite Mean of 3.34 indicates a neutral stance, where users find the environment tolerable but not yet optimal.

This dissatisfaction in the evening is driven by two primary factors: physical congestion and behavioral non-compliance. In terms of crowd density and space, there is a sharp decline from the morning's neutral perception (3.43) to the evening's negative rating (2.62). Specifically, indicators such as "Appropriate Density" (2.58) and "Supports Performance" (2.55) in the evening fall deep into the Disagree range. This confirms that while morning volumes are manageable, the volume of joggers during the 5:00 PM to 8:00 PM window captured by the camera exceeds the comfortable capacity of the track, effectively hindering athletic performance.

Secondly, regarding Crowd Management, the lowest rating in the entire study was observed in "Track Etiquette Followed" during the evening (2.30). This indicates that users feel hindered not just by the number of people, but by a lack of discipline that becomes more acute as density increases. While the communication of rules remained Neutral across both sessions (3.30 and 3.24), the disconnect between knowing the rules and following them suggests a significant gap in active, real-time enforcement. Ultimately, the drop in "Overall Environment" expectations from the morning (3.42) to the evening (2.85) underscores that the current manual monitoring system is insufficient to maintain a consistent, regulated jogging experience throughout the day.

In addition to the Likert scale assessment, joggers provided descriptive responses regarding their perception of crowd levels. These open-ended answers were categorized to determine the dominant sentiment regarding facility congestion across different time slots.

Table 27. Jogger Perception of Crowd Density

Category / Sentiment	Frequency (f)	Representative Responses (Verbatim)
Heavy / Overcrowded	45	<i>"Very Crowded", "Very much crowded", "Very overcrowded", "10 out of 10", "I am not able to jog at my normal pace as a lot of joggers are using the oval"</i>
Crowded / Moderate	36	<i>"Crowded", "Moderately crowded", "Average", "Medium", "Mildly overcrowded", "Somewhat overcrowded"</i>
Light / Manageable	31	<i>"Slightly crowded", "It's not that crowded, depends on the time and day", "Bit crowded but still manageable", "Crowded a little bit", "Its Okay"</i>
Numeric Estimates	4	<i>"uhmm around 50-100", "70", "0.7", "50%"</i>

A thematic analysis of these responses reveals a predominant sentiment of congestion, with 73% of the respondents describing the oval as "Crowded," "Heavy," or "Overcrowded." Specifically, fourteen respondents employed strong intensifiers such as "Very crowded" or "10 out of 10," indicating that the volume of users frequently reaches extreme levels that surpass personal tolerance. Beyond mere density, several participants articulated the functional consequences of this overcrowding, noting that they are "unable to jog at a normal pace" or experience a "surge" of people that disrupts their workout. Only a minority described the conditions as "Manageable" or "Slightly crowded," and these responses were often qualified by specific times, further suggesting that acceptable conditions are the exception rather than the norm.

These qualitative descriptions strongly corroborate the statistical findings presented in SOP 3. The recurring narrative of the oval being Overcrowded explains the low quantitative rating for Item 1 ("Appropriate crowd density"), which received a mean score of 2.58 (Disagree). Similarly, the specific complaints regarding the inability to maintain a jogging pace validate the low rating for Item 6 ("Condition supports consistent performance"). The triangulation of these data points confirms that the "Neutral" rating for physical space is likely misleading if viewed in isolation; while the physical area of the oval may be sufficient, the lived experience of the joggers during peak hours is one of obstruction and constraint, driven by a density that users consistently perceive as excessive.

2.4 Technology and Recommendations

This section addresses the fourth statement of the problem, exploring the feasibility of modernizing the current people-counting system. It presents the respondents' awareness of existing monitoring tools and their willingness to adopt proposed technological interventions, such as real-time crowd alerts and automated occupancy displays. These findings serve as the empirical basis for the recommendations proposed to enhance the efficiency and accuracy of crowd management at the CCSC Oval.

To determine the feasibility of modernizing the people-counting system, the study measured the respondents' awareness of current practices and their willingness to utilize proposed digital solutions. The descriptive statistics are presented in Table 6.

Table 28. Jogger's Awareness and Willingness to Adopt Monitoring Technology

Indicator Statement	Mean ($\bar{x}$)	SD	Verbal Interpretation	
A. Awareness of Current System				
1. I am aware that people counters (cameras/sensors) are used to monitor occupancy.	2.58	1.32	Disagree Neutral	/
2. The people counters used help improve the management of the oval.	3.00	1.30	Neutral	

B. Willingness to Adopt New Technology			
3. I would use alerts or notifications about peak crowd times if they were available.	3.94	1.25	Agree
4. Receiving real-time updates about crowding would help me plan my jogging.	4.03	1.16	Agree
Composite Mean	3.39	Neutral / Agree	

The data reveals a significant visibility gap in the current operations. Item 1 received a low mean of 2.58, indicating that the majority of joggers are unaware that any form of people counting (whether manual or camera-based) is currently taking place. Consequently, Item 2 ("Counters help improve management") received a distinct Neutral rating ($\bar{x} = 3.00$). This suggests that because the counting process is hidden from the public and the data is not shared, users perceive it as having no impact on their experience. The high standard deviation ($S = 1.32$) further indicates confusion or lack of consensus among users regarding what systems are actually in place.

In sharp contrast to the low awareness of current tools, there is a strong and unified demand for user-facing technology. Item 4 ("Real-time updates to plan jogging") received the highest rating in the entire survey ($\bar{x} = 4.03$). This implies that joggers are not passive participants; they are actively seeking data to help them self-regulate their

schedules. Similarly, Item 3 confirmed that users would subscribe to "Crowd Alerts" ($\bar{x} = 3.94$) if offered.

The disparity between Group A (Awareness of Current System: ~2.79) and Group B (Willingness to Adopt: ~3.99) provides a clear empirical mandate for modernization. The data confirms that the jogging community is Technologically Ready. Implementing a system that broadcasts occupancy levels would be immediately adopted by the user base, directly addressing the congestion issues by empowering users to avoid peak hours voluntarily.

To guide the specific selection of technology, the study analyzed the open-ended responses regarding preferred tools. Both joggers and staff were asked: "Which technologies or tools would you prefer or find most helpful?"

Table 29. Stakeholder Preferences for Technological Interventions

Respondent Group	Dominant Preference	Context / Reasoning
Joggers	1. Mobile Notifications Alerts 2. Real-time Occupancy Display	Users prioritized accessibility. They requested “<i>Facebook updates</i>,” “<i>Traffic light system at entrance</i>,” or an “<i>App notification</i>” so they can check crowd levels before leaving home.
Staff	1. AI Cameras / Automated Counters	Staff emphasized operational efficiency. They explicitly requested “<i>AI Camera</i>” to automate the counting process, reducing the burden of manual logging and improving accuracy.

There is a functional alignment between the needs of the staff and the joggers. The staff requires Automated Inputs (AI Cameras) to capture accurate data without manual error, while the joggers require Accessible Outputs (Mobile Alerts/Displays) to make informed decisions. This supports a unified recommendation for an integrated system where an AI Camera (Backend) feeds real-time data to a Public Display or App (Frontend).

The assessment of technological readiness establishes a clear empirical mandate for the modernization of the CCSC Oval's crowd management systems. The data exposes a critical operational disparity: while the current manual counting practices remain largely invisible and undervalued by the public, evidenced by the low awareness and neutral perception of their effectiveness, there exists a robust and unified demand for digital transparency. The overwhelming willingness of respondents to adopt real-time crowd alerts ($\bar{x} = 3.94$) and occupancy updates ($\bar{x} = 4.03$) confirms that the jogging community is not merely receptive to technology but is actively seeking data-driven tools to self-regulate their visitation schedules. Consequently, the transition from passive manual logging to an active, automated monitoring system (such as AI-integrated cameras with public displays) is not only technically feasible but is the most viable intervention to resolve the facility's recurrent congestion. By empowering users with real-time information, the administration can effectively decentralize crowd control, allowing the Evening Peak to naturally dissipate as users voluntarily shift to off-peak hours based on live data.

CHAPTER 3

FINDINGS, CONCLUSION, AND RECOMMENDATION

This chapter synthesizes the data presented in the previous chapter to provide a cohesive analysis of the people-counting practices at the Cebu City Sports Center (CCSC) Oval. It integrates the quantitative results from camera-based headcounts and jogger surveys with the qualitative insights from staff interviews. The discussion is structured to address the four specific research problems: temporal usage patterns, operational challenges, user perception of crowding, and the readiness for technological adoption.

3.1 Findings

The study successfully triangulated data from objective video headcounts, staff interviews, and jogger surveys to reveal critical discrepancies between the facility's theoretical capacity and its actual usage patterns. The following sections detail these findings, supported by data visualizations to illustrate the extent of the congestion and operational limitations.

3.1.1 The Evening Rush Phenomenon and Pattern Synchronization

The investigation into temporal usage revealed that the CCSC Oval operates under a highly synchronized, bi-modal high-density schedule. Contrary to the standard assumption that recreational facilities experience their highest traffic on weekends, the objective headcount data identified Saturday and Sunday as the peak operational days, recording total volumes of 1,474 and 1,432 joggers respectively. This usage trend

suggests that the facility functions primarily as a utility-based fitness hub synchronized with the work and school week, rather than a leisure destination. Within these days, the facility experiences two specific surges: a moderate morning peak occurring between 6:00 AM and 8:00 AM, and a much more intense Evening Rush that spans from 5:00 PM to 9:00 PM. Conversely, the period between 10:00 AM and 3:00 PM represents a dead zone where utilization drops significantly, indicating a major imbalance in how the facility is utilized throughout the day.

A critical discovery regarding these patterns is the accumulation effect that drives the facility's congestion. While the self-reported survey data indicated that the majority of joggers prefer to arrive at 5:00 PM, the physical headcounts revealed that the peak density on the track does not occur until 8:00 PM, where an average of 202 joggers were recorded simultaneously. This discrepancy demonstrates a cumulative density effect. While the initial influx of users begins in the early evening, the objective headcount data reveals that the volume of active joggers continues to rise, reaching saturation at 8:00 PM. This indicates that the rate of users entering the track consistently exceeds the rate of users exiting during this window, causing the crowd to accumulate and peak later than the preferred arrival time.

3.1.2 Systemic Failure of Manual Counting and Data Blind Spots

The assessment of operational procedures confirmed that the current manual counting practices are insufficient for the scale of the facility's operations. Staff respondents rated the accuracy of counting during peak hours with a low weighted mean of 1.88, which corresponds to a verbal interpretation of "Strongly Disagree". The assessment confirmed that during the critical ingress phase between 5:00 PM and 7:00 PM, the volume of entrants moves too fast for personnel to verify tickets individually, forcing them to rely on estimation rather than precise counting. This reliance on visual estimation results in significant variances between reported figures and actual ground-truth occupancy.

Furthermore, a significant data blind spot was identified regarding non-paying users. Interviews with staff revealed that specific groups, such as athletes and students, often bypass the ticketing queue and are therefore excluded from official records. Because the facility relies on ticket sales as the primary metric for attendance, the official data systematically underreports the actual crowd density. This leads to a reactive management style, where staff rely on end-of-shift summaries rather than real-time data, preventing them from proactively closing gates or redirecting flow before overcrowding occurs.

3.1.3 The "Felt Experience" of Overcrowding: Density vs. Compliance

The assessment of user perception validated the quantitative findings, revealing a prevailing dissatisfaction with the current crowd conditions. A substantial 73% of the jogger respondents described the oval as "Crowded," "Heavy," or "Overcrowded," with some utilizing strong descriptors such as "10 out of 10" to characterize the density . This sentiment was statistically corroborated by the survey results, where the appropriateness of crowd density received a negative mean rating of 2.58. Users reported that the current conditions prevent them from maintaining consistent jogging performance, effectively transforming the workout environment into an obstacle course where they must constantly break stride .

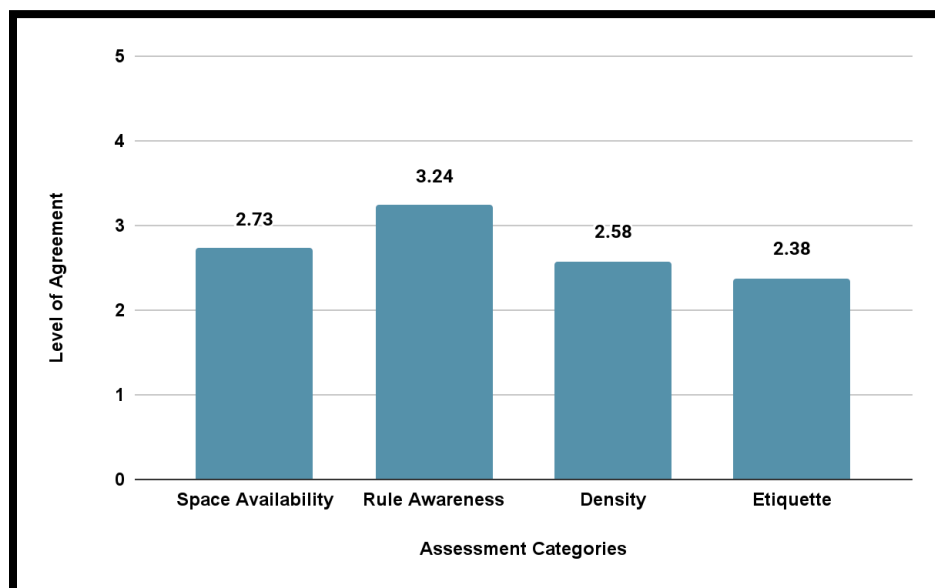


Figure 15. User Perception of Crowd Conditions (5-Point Scale)

Beyond the physical number of people, the study identified that the sensation of congestion is exacerbated by a lack of rule compliance. The lowest rating in the entire

study was given to track etiquette, with a mean score of 2.30. This indicates that users feel hindered not just by the volume of people, but by the lack of lane discipline among them. The disparity between "Rule Awareness" (Mean = 3.24) and "Rule Compliance" (Mean = 2.30) suggests that while users know the rules, they do not follow them, largely due to a lack of active enforcement or marshaling on the track itself .

3.1.4 Technological Readiness and the Demand for Transparency

The study established a high level of technological readiness among both staff and joggers, highlighting a clear mandate for modernization. There is currently a visibility gap in operations, as most joggers are unaware that any form of monitoring is taking place, resulting in a low awareness rating of 2.58. This lack of visibility leads users to perceive the management as passive or uninvolved in crowd control.

However, there is a robust demand for transparency and data-driven tools. The jogging community expressed a strong willingness to adopt digital solutions, with respondents agreeing that they would use real-time occupancy updates (Mean = 4.03) and crowd alerts (Mean = 3.94) to help them plan their visits. This alignment between the staff's need for automated counting tools and the users' desire for accessible information indicates a favorable environment for implementing the proposed technological interventions. The users are not merely passive participants; they are actively seeking data to help them self-regulate their schedules and avoid peak hours .

3.2 Conclusion

The comprehensive assessment of the people-counting practices at the Cebu City Sports Center (CCSC) Oval leads to the conclusion that the existing manual system is fundamentally inadequate for managing the complexities of a high-density urban athletic facility. Through the triangulation of headcount data and user surveys, the study confirms that facility usage follows a highly synchronized, bi-modal temporal pattern. The critical surges identified during weekday Evening Rushes are not merely anecdotal but represent a consistent operational strain that the current manual infrastructure is not equipped to handle.

A primary conclusion of this research is the existence of a systematic Data Blind Spot. Because the facility relies almost exclusively on ticket sales logs as its metric for occupancy, it fails to account for a significant portion of the actual population, including student-athletes and city-sponsored groups. This results in official records that consistently underreport the physical load on the track. This lack of data precision is dangerously compounded by an Accumulation Effect. The study concludes that overcrowding is driven less by late-night arrivals and more by the overlapping dwell times of joggers who arrive early and remain on the track. By the 8:00 PM saturation point, the manual visual estimation method used by staff becomes completely fallible, leaving management without the real-time figures necessary to make critical safety decisions, such as temporary gate closures or lane redirections.

Table 30. Justification for Operational and Data Conclusions

Justification Pillar	Research Evidence/Finding	Logical Conclusion
Accuracy Gap	Staff rated manual counting accuracy at 1.88 (Strongly Disagree).	Manual tallying fails during high-ingress surges (5 PM - 7 PM).
Inclusivity Gap	Non-paying users bypass the ticketing-based recording system.	Official logs systematically underreport the true physical load of the track.
Time Gap	Peak density (202 joggers) occurs at 8:00 PM, long after ticket sales peak.	Managing entry volume is not enough; "dwell time" must be monitored.

Furthermore, the research highlights a significant "Perception-Compliance Gap." The conclusion drawn from user feedback is that the oval has transitioned from a high-performance fitness venue into a hazardous "obstacle course." The data suggests that while users are aware of the rules, the absence of a visible, automated monitoring system creates a sense of lawlessness on the track. Without objective, real-time feedback on crowd density, users have little incentive to maintain lane discipline or etiquette. This lack of transparency results in a degraded "felt experience" where 73% of users feel their athletic performance is actively hampered by the environment, turning a public service into a source of frustration.

Table 31. Justification for User Perception Conclusions

Justification Pillar	Research Evidence/Finding	Logical Conclusion
Quality of Service	73% of joggers perceive the oval as "Heavy" or "Overcrowded."	High density directly hampers consistent athletic performance.
Compliance Gap	Etiquette and Lane Discipline received the lowest mean score of 2.30.	Awareness of rules does not translate to compliance without active monitoring.
Monitoring Gap	Low user awareness (2.58) of any existing crowd control measures.	Users perceive management as passive, leading to a breakdown in track order.

Finally, the study concludes that there is a robust mandate for modernization. There is a clear alignment between the operational struggles of the staff and the digital expectations of the public. Joggers are no longer satisfied with being passive participants; they are actively seeking data-driven tools, such as real-time crowd alerts, to help them self-regulate their own schedules. Therefore, it is the final conclusion of this study that the transition from manual logging to an AI-integrated, automated monitoring system is not merely a technological upgrade, but a necessary administrative shift. Implementing such a system is the only viable path to eliminating data inaccuracies, ensuring public safety, and restoring the CCSC Oval to its intended functional purpose as a premier athletic destination.

Table 32. Justification for Technological Readiness Conclusions

Justification Pillar	Research Evidence/Finding	Logical Conclusion
Tech Readiness	High willingness to use Real-Time Updates (4.03) and Crowd Alerts (3.94).	The public is ready to adopt digital solutions to avoid peak hours.
Operational Need	Staff confirmed the inability to verify tickets individually during rushes.	Automated, AI-integrated counting is the only way to achieve 24/7 accuracy.

In conclusion, the study asserts that transitioning from passive manual logging to an active, AI-integrated digital monitoring system is an essential administrative evolution rather than a simple technological upgrade. By shifting from a culture of 'estimated management' to a framework of 'data-driven governance,' the CCSC can resolve persistent data inaccuracies and fundamentally bolster public safety. This modernization represents the only definitive path toward restoring the oval's integrity as a premier, safe, and systematically organized athletic destination for the Cebuano community.

3.3 Recommendation

In light of the conclusions drawn, which identified critical gaps in manual tallying and confirmed user dissatisfaction with overcrowding, the researchers propose the following recommendations. To ensure practical application, the specific action plans are presented below, categorized by the issue addressed.

3.3.1 Technological Recommendations

Table 33. Recommendation on Data Accuracy

Identified Issue	Recommended Action	Responsible Unit	Expected Outcome
Manual tallying fails to capture students and athletes.	Deploy cameras at the oval for real-time, automated data capture.	CCSC Administration & IT Department	Accurate, proactive management with visibility of all user groups.

The primary recommendation is for the Cebu City Sports Center administration to transition from manual tallying to an automated people-counting system. Specifically, the implementation of AI-powered cameras at the CCSC oval area is advised. This technology addresses the Visibility Gap by ensuring accurate, real-time data capture that includes currently uncounted groups like students and athletes. This system should be integrated into a central dashboard to provide security personnel with immediate visibility into crowd density, allowing for proactive rather than reactive management decisions.

Table 34. Recommendation on User Transparency

Identified Issue	Recommended Action	Responsible Unit	Expected Outcome
Joggers lack information to avoid overcrowding.	Display real-time occupancy on a public monitor or mobile platform.	IT Department & Public Relations	Empowered users who can voluntarily avoid peak hours.

Furthermore, to address the user demand for transparency regarding space availability, the study recommends the development of a "CCSC Crowd Alert" mechanism. This solution involves displaying real-time occupancy levels via a public monitor at the entrance or through a mobile-accessible platform. By making this data public, the administration enables joggers to make informed decisions and voluntarily avoid peak hours, thereby flattening the demand curve without requiring strict entry bans.

3.3.2 Policy and Operational Recommendations

Table 35. Recommendation on Functional Congestion

Identified Issue	Recommended Action	Responsible Unit	Expected Outcome
Functional Congestion	Enforce Lane Discipline & Zoning		
Poor etiquette and lack of discipline cause friction.	Implement clearer zoning for fast/slow runners and marshal deployment.	Security Personnel & Sports Management	Smooth traffic flow and maximized usable track space.

Technological solutions must be supported by operational changes to mitigate the functional congestion. While the Technical Recommendation (the automated counting system) serves as the diagnostic tool that provides real-time occupancy data, this Policy Recommendation serves as the operational response mechanism.

To answer the specific needs of scheduling, it is recommended that the facility adopt a Data-Driven Enforcement Policy. Instead of implementing static rules that may be ignored, management should utilize the historical and real-time data generated by the proposed system to decide the Scheduling Policy:

- **Dynamic Marshal Deployment:** The system identifies specific "Red Zone" hours (e.g., Tuesday at 8:00 PM, as found in the study) where occupancy exceeds 100%. Marshals should be scheduled specifically during these data-verified peaks to enforce lane discipline, ensuring staff resources are not wasted during low-volume hours.
- **Adaptive Zoning:** When the system alerts that the Functional Inner-Lane Capacity (181 joggers) is reached, a "Congestion Protocol" should be triggered where outer lanes are strictly designated for walkers to decompress the inner track.

By linking the technical output directly to the operational policy, the CCSC can transition from reactive crowd control to proactive, evidence-based management.

Table 36. Recommendation on Institutional Overcrowding

Identified Issue	Recommended Action	Responsible Unit	Expected Outcome
Institutional Overcrowding	Implement Group Scheduling Policy	CCSC Admin & School Coordinators	Flattened demand curve and reduced density during public peak hours.
Large student groups coincide with public peak hours.	Restrict large P.E. classes/teams to off-peak hours (9:00 AM – 3:00 PM).		

Additionally, to manage the physical volume of users, the administration should implement a Scheduling Policy for Large Groups. The findings indicate that the influx of students contributes significantly to the peak hour density. To mitigate this, large P.E. classes or sports teams should be restricted from using the oval during the public peak hours (5:00 PM – 8:00 PM) and instead be scheduled during low-traffic windows (e.g., 9:00 AM – 3:00 PM). This policy would redistribute the crowd load, preventing the facility from exceeding its comfortable carrying capacity.

Table 37. Recommendation on Peak Intensity Safety

Identified Issue	Recommended Action	Responsible Unit	Expected Outcome
Peak Window Intensity	Targeted Marshal Deployment	Security Personnel	Enhanced safety during the most critical high-density periods.
High volume peaks confirmed on weekends.	Deploy marshals specifically during the critical Friday to Sunday evening window.		

Crucially, policy enforcement must be targeted where it is needed most. The administration should prioritize the deployment of lane marshals during the critical Friday to Sunday evening window, which the study identified as the period of highest intensity. These marshals would ensure smooth flow and maximize the usable space of the track by actively correcting non-compliant behavior, ensuring that the physical capacity of the oval is utilized efficiently.

3.3.3 Recommendations for Future Research

To further validate and expand upon the findings of this study, it is recommended that future researchers address the limitations regarding the timeframe and financial scope of the proposed solutions. First, a Longitudinal Analysis should be conducted over a period of 3 to 6 months. This extended duration is crucial to capture significant seasonal variations, such as the high-traffic weeks leading up to major events like CVIRAA or Sinulog. Staff members noted these as "Super Peak" times, and a study limited to one week may not fully represent the maximum crowd density the facility experiences. Second, a Cost-Benefit Analysis should be performed to assess the economic viability of the technological interventions. Future studies should conduct a feasibility comparison between specific solutions, such as Turnstiles versus AI Cameras, to provide the CCSC management with a concrete financial roadmap for implementing the recommended modernization.

ETHICAL CONSIDERATIONS

The researchers uphold the integrity of the study by ensuring that all participants, the community, and the research environment are protected, and that all procedures are conducted with responsibility and transparency. Guided by the principles of respect for persons, beneficence, and justice, the study recognizes participants as autonomous individuals who have the right to make informed decisions about their involvement. Respect for persons was observed by obtaining informed consent from all CCSC staff and jogger respondents prior to interviews and survey participation, ensuring they understood the study's purpose, procedures, and their right to withdraw at any point. Beneficence was maintained by minimizing risks through non-intrusive data collection methods such as systematic observation in public areas, anonymous surveys, and structured interviews that avoided sensitive personal information. Justice was ensured by treating all participants fairly and providing equal opportunity to contribute their insights regarding current people-counting practices. As part of ethical compliance, the study will secure approval from the Recoletos Ethics Review Office (RERO) before data collection. All collected data will be handled in accordance with the Data Privacy Act of 2012, with responses anonymized, securely stored, and used solely for academic research purposes. Through these measures, the study ensures ethical conduct while generating meaningful insights into improving people-counting practices at the Cebu City Sports Center Oval.

A. Documentation of Informed Consent

The documentation of informed consent in this study will ensure that all participants fully

understand their rights, the study procedures, and the voluntary nature of their involvement before any data is collected. During the face-to-face orientation, participants will receive the Informed Consent Form (ICF) either in printed form or digitally through Google Forms, depending on their preference. The ICF will clearly outline the study's purpose, methodology, potential risks and benefits, confidentiality measures, and data protection

procedures. Participants will be given sufficient time to read and review the document, and they will be encouraged to ask questions or seek clarification. The researchers will remain available throughout the process to address concerns and support informed decision-making.

For digital consent, participants will be provided with a device—such as a tablet or laptop—to access the ICF and complete the questionnaire through Google Forms. The link to the digital ICF will also be emailed to them, allowing completion at their own convenience. Consent will be documented electronically by clicking “Agree,” and providing their email, which will serve as their digital signature. Selecting “Disagree” will automatically end the process, and no data will be collected. Participants who provide online consent will receive an emailed copy of the ICF immediately after the session for their personal reference.

For print consent, participants will sign a physical copy of the ICF during the in-person meeting. The researcher will then create a secure digital copy by photographing the signed form, which will be used solely for documentation and audit purposes. The

original signed copy will be returned to the participant for their personal records. Printed materials, such as survey questionnaires, will be placed in individual sealed envelopes and stored in a locked cabinet to maintain confidentiality. Access to these physical documents will be restricted to the authorized researcher only.

To uphold strict confidentiality and comply with ethical standards, all collected data—including digital and scanned documents—will be anonymized and stored securely in encrypted folders on Google Drive. Access will be limited only to authorized members of the research team. These procedures ensure that informed consent is properly documented while protecting the rights, privacy, and welfare of all study participants.

B. Recruitment and Selection of Participants

The recruitment and selection of participants were carried out through an ethical and systematic process to ensure that both key facility personnel and regular users of the Cebu City Sports Center (CCSC) Oval were appropriately represented in the study. A total of 13 CCSC staff members—including ticketing officers, security personnel, and office staff responsible for visitor logs—were selected through purposive sampling because their roles provide direct insight into daily operational procedures and existing people-counting practices. For the survey component, 65 jogger participants were recruited through convenience sampling during their actual visit to the oval to ensure that their responses reflected real-time experiences and perceptions related to crowd monitoring and facility usage.

To guide the selection process, the study established clear inclusion criteria for jogger respondents. Eligible participants were required to (1) be active users of the CCSC oval during the scheduled data collection period, (2) hold a valid jogger's pass, (3) be 18 years old or above, and (4) voluntarily provide informed consent. Participants also needed to be capable of completing either the printed or digital version of the survey instrument. As part of the study's observational protocol, a research-approved camera was installed at the main entrance and exit—authorized by the CCSC administrator—to document user flow patterns. Posters were prominently displayed to inform the public that the camera was installed strictly for research purposes.

C. Incentives and Compensation

Participants in the survey will not receive any incentives or monetary compensation, ensuring that their involvement in the study remains entirely voluntary and free from any form of undue influence. This approach upholds ethical standards by preventing coercion and maintaining the integrity of participant decision-making. Although no material benefits will be provided, the researcher will express appreciation by sending a personalized message to each participant, acknowledging their time, cooperation, and valuable contribution to the study.

D. Community Consideration

The study was conducted with full awareness of the unique characteristics and

sensitivities of the community that regularly uses the Cebu City Sports Center Oval. Throughout the research process, the researchers ensured that all procedures—such as on-site recruitment, observations, and the installation of a research-approved camera—were carried out in a manner that was considerate of the daily routines and expectations of facility users. Clear informational posters were placed at entry and exit points to maintain transparency and avoid any discomfort or misunderstanding among joggers and staff. Interactions with participants were conducted respectfully, ensuring that individuals had the freedom to decline participation without pressure. The data collection schedule was also designed to avoid disrupting regular activities, particularly during peak usage hours.

E. Vulnerability of Participants

The survey participants in this study are considered non-vulnerable because they are all adults who are 18 years old and above, meeting the legal age required to understand the study procedures and provide informed consent independently. Individuals within this age group possess the cognitive capacity to evaluate the information presented, ask questions, and voluntarily decide whether to participate. Completing the survey poses no physical, emotional, or social risk beyond their usual activities at the CCSC facility, ensuring their safety throughout the process. Participation is entirely voluntary, and no incentives are offered, preventing any form of pressure or undue influence.

F. Benefits to the Participants

Participation in this study offers indirect benefits to respondents by giving them an opportunity to reflect on their own experiences using the CCSC oval and their observations of crowd conditions and monitoring practices. Although the study is not expected to provide direct psychological or material benefits to participants, their input can contribute to improvements in the facility's people-counting system, which may lead to a safer and more organized jogging environment in the future. Sharing their perspectives, participants help inform recommendations that can enhance crowd management and overall user experience at the CCSC. Indirectly, this may lead to long-term benefits for the community of joggers and facility staff.

G. Potential Risk and Safety of the Participants

The study poses minimal risk to participants because all activities involve only surveys, brief interactions, and non-intrusive observation in a public setting. No physical, psychological, or social harm is expected, and participants will not be asked to provide sensitive or personal information. The installation of a research-approved camera at entry and exit points will capture only general user movement, not individual identities, ensuring safety and privacy. Participation is completely voluntary, and individuals may withdraw at any time without consequence.

H. Privacy and Confidentiality

The privacy and confidentiality of all participants will be strictly protected throughout the

study. Survey respondents will remain anonymous, and no personal identifiers or sensitive information will be collected. The camera installed at the entrance and exit will capture only the entry and exit times of users to support the observation of crowd flow, without recording identifiable facial features or linking data to specific individuals. All survey responses and observational data will be stored securely in encrypted, password-protected files accessible only to the researcher. Information obtained from the study will be used solely for research purposes and reported in aggregate form to prevent any individual from being identified.

I. Debriefing and Referrals

After completing the survey, participants will undergo a brief debriefing to ensure that they feel comfortable and informed about their involvement in the study. For those who complete the printed survey, the researcher will gently check in with them afterward to ask about their experience, clarify any parts of the study they may have found unclear, and ensure that they did not feel discomfort or pressure while responding. Participants will be encouraged to share any comments or concerns about the process. For those completing the digital survey, a short post-evaluation section is integrated into the Google Forms to allow participants to express their thoughts or feelings about their survey experience immediately after submission. Although the study poses minimal risk, participants may contact the researcher at any time should questions arise, and referrals or guidance will be provided if anyone experiences unexpected discomfort related to their participation.

J. Data Storage, Retention and Destruction

All data collected in this study—including anonymized survey responses, observational records, and documented entry and exit times—will be stored securely in encrypted, password-protected folders on Google Drive accessible only to the researcher. Printed materials, such as completed paper surveys, will be kept in sealed envelopes and locked in a secure cabinet. Data will be retained for one year after the completion of the study to allow sufficient time for validation, analysis verification, and possible academic requirements such as panel review, presentation, or publication. After the retention period, all digital files will be permanently deleted from storage, and printed documents will be shredded to ensure complete destruction.

K. Conflict of interest

The researcher declares that there is no conflict of interest in the conduct of this study. The researcher has no financial, professional, or personal relationships with the Cebu City Sports Center, its administrators, staff, or joggers that could influence the study's outcomes. All data collection and analysis procedures were carried out independently and objectively, ensuring that the results are free from bias. The sole purpose of the study is to contribute to academic understanding and improve people-counting practices at the facility, without any expectation of personal gain.

L. Qualification of the Researcher

The researchers are undergraduate students enrolled in the Bachelor of Science in

Computer Engineering (BSCpE) program, which has equipped them with relevant academic preparation for conducting this study. Through their coursework, they have completed subjects such as Research Methods, Statistics, Data Analysis, Systems Analysis and Design, Computer Architecture, and courses related to data processing and system monitoring—providing them with the technical and methodological background necessary for assessing people-counting practices. They have also participated in institutional seminars and training sessions on research methodology and research ethics, enhancing their competence in designing ethical studies, protecting participant confidentiality, and applying appropriate analytical tools. As student researchers, they acknowledge the limitations of their experience, particularly in advanced methodological decision-making and technical validation. To ensure academic rigor and ethical compliance, they conducted the study under the close supervision of their research adviser and with guidance from faculty consultants when needed. This support ensured that all research activities were carried out responsibly, ethically, and in alignment with institutional and academic standards.

DEFINITION OF TERMS

Overcrowding – A condition where the number of joggers present in the CCSC oval exceeds the estimated maximum lane-based or overall capacity (approximately 1,600 joggers). In this study, overcrowding is defined as occupancy rates exceeding 100% of the oval's carrying capacity, leading to safety concerns, congestion, and potential limitations on jogger mobility.

Attendance – Refers to the total number of joggers present within the CCSC oval during a specific time period. In this study, attendance is measured through manual counts, personnel records, and crowd monitoring practices to determine usage trends and peak periods.

Congestion Hotspot – Specific areas within the CCSC oval where crowd density becomes high enough to restrict the smooth movement of joggers. In this study, congestion hotspots are identified using foot traffic patterns, occupancy rates, and observed flow disruptions.

Cronbach's Alpha (α) – A statistical measure of internal consistency or reliability of grouped questionnaire items. A coefficient of 0.70 or higher indicates that the items reliably measure the same construct. This study employs Cronbach's Alpha to test the reliability of Likert-scale responses in the jogger survey.

Crowd Monitoring – Refers to the systematic observation and tracking of jogger movement and attendance within the CCSC oval. In this study, crowd monitoring involves manual counts, personnel observations, and documented records to assess turnout, identify congestion patterns, and ensure occupancy levels remain within safe limits.

Frequency Distribution – A statistical method that organizes data into categories showing how often each response occurs. In this study, frequency distribution is used to summarize jogger turnout counts and survey responses.

Jogger Turnout – The number of joggers who enter and use the CCSC oval within a given time interval. It is measured in daily, hourly, and 10-minute intervals to assess peak usage periods and occupancy.

People Counting Practices – Refers to the methods, tools, and processes employed to monitor and record the number of joggers entering and using the CCSC oval. This includes ticketing records, manual documentation, and personnel monitoring.

Peak Hours / Peak Periods – Time intervals when jogger turnout reaches its highest levels at the CCSC oval. In this study, peak periods are identified through observation and manual counts summarized by daily, hourly, and 10-minute intervals.

Percentage Distribution – A statistical tool used to express the proportion of responses or turnout in relation to the total number of respondents or joggers, providing comparative insights across categories.

Standard Deviation (S) – A statistical measure that shows the variability of responses in relation to the mean. In this study, it is used to determine whether jogger assessments in the questionnaire are consistent or widely dispersed.

Thematic Coding – A qualitative analysis technique involving the process of categorizing textual data into codes and themes. This study uses thematic coding to analyze responses from CCSC personnel interviews regarding monitoring and people-counting practices.

Turnout Distribution – Refers to the spread of jogger attendance across different time intervals within the day. In this study, the distribution of turnout helps determine peak usage and assess whether occupancy limits are exceeded.

Evening Rush - refers to the primary daily window of peak facility utilization, specifically occurring between 5:00 PM and 9:00 PM.

Data Blind Spot - refers to the systematic exclusion of specific non-paying user groups from the official facility attendance records.

Dead Zone - refers to the five-hour midday block, typically between 10:00 AM and 3:00 PM, characterized by a precipitous drop in facility utilization. This period represents a significant temporal inefficiency where activity often falls to single-digit counts, contrasting the high-density surges observed during the morning and evening peaks.

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APPENDIX/APPENDICES

INFORMED CONSENT FORM (ICF) – PRINTED FORM FOR

Survey Respondents

Dear Participants We are students from the University of San Jose–Recoletos (USJ-R), Cebu City, currently enrolled in the Bachelor of Science in Computer Engineering (BSCpE) program, which requires the completion of an undergraduate research study

as part of our academic requirements. Our research study, “Assessing the Current People Counting Practice at the Cebu City Sports Center Oval,” is being conducted by Eric Mar A. Geonzon, Chashmyr Ignacio, and Mykylla Angelyka O. Jordan under the guidance of our assigned faculty adviser. The general objective of this study is to evaluate the effectiveness of the existing people-counting system at the CCSC oval and identify areas for improvement in crowd monitoring and data accuracy.

This Informed Consent Form (ICF) explains the purpose of the study, what your participation involves, the possible risks and benefits, and how your information will be protected. Please read and understand this ICF carefully before deciding whether to participate.

Your participation is entirely voluntary, and you have the right to refuse or withdraw from the study at any time without penalty or negative consequence.

We sincerely appreciate your time and consideration. Your insights and experiences as a jogger using the CCSC oval are important and will help us better understand the current people-counting practices and how they can be improved for the safety and convenience of all facility users.

A. Overview and Purpose of the Study

This study explores how joggers enter, exit, and use the Cebu City Sports Center (CCSC) oval, along with how staff currently monitor and count daily users. The purpose of the research is to evaluate the effectiveness of the existing people-counting system

and identify possible improvements that can help make crowd monitoring more accurate and efficient. Your participation will provide valuable insights that can support better management and a safer experience for all joggers using the facility.

B. Study Procedure

If you agree to this Informed Consent Form (ICF), you will be asked to answer a survey either online through Google Forms or using a printed questionnaire, depending on what you prefer. This study needs 65 jogger participants, based on the number of people who regularly use the CCSC oval. Before you begin, you must meet the following requirements: (1) you are 18 years old or older, (2) you are an active user of the CCSC oval, and (3) you have a valid jogger's pass. If you do not meet these requirements, you cannot continue with the survey.

If you choose the online survey, you will click "I Agree" and be directed to Google Forms. If you choose the printed survey, the researcher will give you a paper questionnaire with simple instructions. Both versions will ask about your jogging experience, crowd conditions, and how people are counted at the CCSC oval. The survey takes about 10–15 minutes to finish.

Joining the study is voluntary. You may skip any question or stop at any time without any problem. For online participants, a short section at the end will ask how you felt while answering the survey. For printed surveys, the researcher will check on you briefly after you finish to make sure you are comfortable and to answer any questions.

C. Potential Risk and Discomfort

You may experience minor discomfort while answering the survey, such as feeling tired or unsure about some questions. There are no expected physical, emotional, or social risks from joining this study. You may skip any question or take a break at any time to reduce any discomfort.

D. Potential Benefits to the Participants and Society

There may be no direct benefits to you, but answering the survey may help you reflect on your experience as a jogger at the CCSC oval. Your participation will contribute to improving the people-counting system and overall safety and organization at the facility. The results of the study may benefit the community by helping the CCSC provide better services for all users.

E. Privacy and Confidentiality

Your identity will remain anonymous, and no personal information such as your name will be collected. All your answers will be kept private and stored securely, accessible only to the researcher. Information from the survey will be reported in general form so no one will know who provided specific responses.

F. Participation and Withdrawal

Your participation in this study is completely voluntary. You may stop at any time or choose not to answer a question without any penalty—just inform the researcher if you decide to withdraw. Whatever decision you make will be respected, and there will be no negative consequences.

G. Incentives and Compensation

You will not receive any money or material rewards for joining this study. The researchers will, however, give you a short thank-you message to show appreciation for the time and effort you shared in completing the survey.

H. Rights of the Participants

As a participant in this survey, you have the right to:

- Decide whether or not you want to join the study. Your participation is completely voluntary.
- Stop answering the survey at any time. If you decide to withdraw, please inform the researcher so your responses will not be included.
- Skip any question that makes you feel uncomfortable or that you prefer not to answer.
- Understand the purpose of the study, what you will be asked to do, and how your data will be used before you agree to participate.
- Remain anonymous; no personal information that can identify you will be collected.

- Have your answers kept secure and accessed only by the researcher. Your responses will be reported in summary form so no one can identify you.
- Ask the researcher for clarification before, during, or after the survey.

I. Monitoring and Oversight

This study will be supervised by the research adviser and monitored by the Recoletos Ethics Review Office (RERO) to ensure that all ethical standards and guidelines are followed throughout the research process.

J. Qualification of the Researchers

The researchers are Bachelor of Science in Computer Engineering (BSCpE) students trained in basic research methods, data gathering, and ethical procedures. They have completed coursework related to data analysis, systems monitoring, and research methodology, which supports their ability to conduct this study responsibly. A faculty adviser and academic consultants will guide and assist the researchers to ensure accuracy, proper conduct, and ethical compliance.

K. Investigator's Contact Information

If you have any questions or concerns about this study, you may contact Chashmyr Ignacio through 09564531316 or email at chashmyr.ignacio.22@usjr.edu.ph. If you prefer to speak with someone independent of the research team, you may contact the

Recoletos Ethics Review Office (RERO) at telephone number (032) 253-7900 local 336
or email usjrerb@usjr.edu.ph

PART II: PARTICIPANTS CONSENT

A. Certificate of Research Participants Consent.

I have read the foregoing information and/or it has been read to me. I have had the opportunity to ask questions about the study protocol. My questions have been answered to my satisfaction, and I agree and consent voluntarily to participate in this study. I can withdraw my consent at any time and discontinue participation without penalty.

Signature above printed name of participants

Date Signed

B. Statement by the researcher or person taking consent

I have accurately read out and /or explained the research protocol to the potential participant, and to the best of my ability made sure that he/she understands the information described in the research protocol in this document. Moreover, I confirm that the participants were given an opportunity to ask questions about the study protocol and all the questions asked by the participant have been answered correctly to the best of

my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily. A copy of this Informed Consent Form has been provided to the participants.

Name of Researcher Obtaining Consent
Printed and Signed

Date Signed

INFORMED CONSENT FORM (ICF) – Google Form

For Survey Respondents

We are pleased to invite you to participate in our study titled “Assessing the Current People Counting Practice at the Cebu City Sports Center Oval.” This research is conducted by Eric Mar A. Geonzon, Chashmyr Ignacio, and Mykylla Angelyka O. Jordan, BS Computer Engineering students at the University of San Jose–Recoletos. Your insights and experiences are invaluable, and your participation will make a significant contribution to the success of this research.

This consent form provides essential information about your participation. It outlines the study’s objectives, the procedures involved, potential risks and benefits, and how your privacy will be protected. Please note that your participation is entirely voluntary. You are free to decide whether or not to participate, and you may withdraw from the study at any time without penalty or loss of benefits.

Thank you for considering participation in this important research. Your involvement will play a key role in advancing our understanding of the current people-counting practices and crowd monitoring accuracy at the CCSC Oval.

A. Purpose of the Study

The purpose of this study is to evaluate the current people-counting practice at the Cebu City Sports Center (CCSC) Oval using a Google Form survey to gather experiences and feedback from joggers and staff.

B. Study Procedure

If you choose to participate in this study, you will first be asked to review and sign an informed consent form. This form will provide you with essential details about the study, including its purpose, procedures, potential risks and benefits, and the measures in place to protect your privacy.

Once you provide your consent, you will complete a survey questionnaire titled “Survey on Current People Counting Practices at the CCSC Oval.” The first phase of the study involves gathering data through this survey, which focuses on feedback from joggers and staff on current people-counting accuracy, crowd monitoring, and peak-hour patterns. The survey will take approximately 15–20 minutes to complete, and your responses will remain confidential.

After completing the survey, you may be invited to participate in a follow-up interview to provide deeper insights into your experiences and perspectives. The interview will be conducted by the researcher and will last between 30 minutes to one hour. With your

permission, the interview will be audio recorded to ensure accurate data collection. You will be informed of this recording beforehand, and it will be used solely for the purpose of this study. Participation in the interview is entirely voluntary, and you may decline to be recorded or withdraw from the study at any time without penalty.

It is important that you answer all questions in the survey and interview honestly and to the best of your ability, as your truthful responses are essential to the accuracy of the study's findings. You are free to skip any questions you are uncomfortable answering, and you may withdraw from the study at any time without penalty.

C. Potential Risk and Discomfort

There are minimal potential risks associated with participating in this study, such as temporary discomfort or inconvenience while completing the survey. To ensure your safety, all responses will remain confidential, and any identifying information will be securely protected throughout the research process.

D. Potential Benefits to the Participants and Society

Participants in this study may gain valuable insights into the current crowd-monitoring and people-counting system at the CCSC Oval, which may enhance their understanding of how attendance tracking is managed. Additionally, their involvement will contribute to advancing knowledge that could improve people-counting accuracy and overall crowd management that may benefit the community or society at large.

E. Privacy and Confidentiality

Your privacy and confidentiality are of utmost importance in this study. All data collected will be anonymized to ensure that no personally identifiable information is linked to your responses. The information you provide will be securely stored, accessible only to authorized researchers, and used solely for the purposes of this research.

F. Participation and Withdrawal

Participation in this study is completely voluntary, and you are free to choose whether or not to take part. You may withdraw from the study at any time, without any penalty or loss of benefits. If you decide to withdraw, your data will be excluded from the study, and you will not be required to provide any further information.

G. Incentives and Compensation

Participants will not receive any incentives, but the researcher will send a personalized thank-you message to acknowledge and appreciate their time and effort in the study.

H. Rights of the Participants

As a participant in this research study, you are afforded certain rights to ensure your comfort, safety, and informed involvement. It is important to us that you understand these rights fully:

- You have the right to receive all relevant information about the study before deciding whether to participate.
- Your personal information and responses will be kept confidential and will not be shared with anyone outside of the research team.

- You have the right to withdraw from the study at any point, for any reason, without any negative consequences or impact on your relationship with the researchers.
- You have the right to ask questions about the study at any time before, during, or after your participation, and the researchers will provide prompt and clear responses.

I. Monitoring and Oversight

This study will be overseen by the research adviser of the research team and the Research Ethics Committee of the Recoletos Ethics Review Office (RERO).

J. Qualification of the Researcher

The researchers are college students majoring in Computer Engineering, with relevant academic training and coursework to conduct the study. The research adviser, along with experts in the field, will provide guidance and oversight to ensure the study's methodology and analysis are sound and adhere to ethical standards.

K. Investigator's Contact Information

If you have any questions or concerns about the research, please feel free to contact:

Chashmyr Ignacio

Mobile: **0956-453-1316**

Email: chashmyr.ignacio.22@usjr.edu.ph

If you want to talk to someone independent of the research team, please contact the **Recoletos Ethics Review Office (RERO)** through telephone **253-7900 local 336** or email usjrerb@usjr.edu.ph.

PART II: PARTICIPANTS CONSENT

A. Certificate of Research Participants Consent.

I have carefully read the information provided above regarding this study, or it has been presented to me in an understandable format. I have been given the chance to ask questions about the study's protocol and procedures. My questions have been addressed satisfactorily, and I voluntarily agree to participate in this study. I acknowledge that I can withdraw my consent and discontinue my participation at any time without any penalty or negative consequences

☐ I have read and agree to the above statement. I consent to voluntarily participate in the study.

☐ I have read and do not agree to the above statement. I do not consent to participate in the study.

Email of the Participants _____

NOTE: The email is an alternative to e-signature of the participants to prevent data privacy violation. This will also be used only when there are concerns needed to address with regards to your participation in the study.

Google form: <https://forms.gle/bxayPD8sphNirqDV7>

Survey Questionnaire

I. Questionnaire

Crowd Monitoring Personnel Questionnaire

1. What are the usual peak hours during the day in your assigned area?
 - ☐ Before 6 AM
 - ☐ 6 AM to 9 AM
 - ☐ 9 AM to 12 PM
 - ☐ 12 PM to 3 PM
 - ☐ 3 PM to 6 PM
 - ☐ After 6 PM
2. Which days of the week experience the highest volumes of visitors?
 - ☐ Monday to Friday (weekdays)
 - ☐ Saturday
 - ☐ Sunday
 - ☐ No significant difference
3. Have you encountered an emergency related to crowding or congestion in your area?
 - ☐ Yes
 - ☐ No
4. If yes, what type of emergency occurred most frequently? (Select all that apply)
 - ☐ Medical emergency (injury, illness)
 - ☐ Fire or hazard

- ☐ Crowd stampede or panic
- ☐ Equipment or infrastructure failure
- ☐ Other (please specify)_____

5. What criteria do you use to identify or classify crowd-related incidents? (Select all that apply)

- ☐ Number of people exceeding a safe limit
- ☐ Difficulty in movement or flow
- ☐ Reports of safety hazards
- ☐ Complaints from visitors
- ☐ Detection by monitoring technology
- ☐ Other (please specify)_____

6. Are you provided with the necessary equipment (e.g., radios, flashlights, access to monitoring systems) for effective crowd management?

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

7. Which methods do you use for monitoring crowd size and movement? (Select all that apply)

- ☐ Visual observation by assigned personnel

- ☐ Counting entrants manually
- ☐ Communication with other staff for updates
- ☐ Use of whistles, flags, or signs for signaling
- ☐ Verbal reports from visitors

8. What is the average number of staff assigned to monitor the crowd during peak periods?

- ☐ 1-2
- ☐ 3-5
- ☐ 6-8
- ☐ More than 8

9. How do you communicate crowd condition updates with other team members?

- ☐ Radios or walkie-talkies
- ☐ Face-to-face communication
- ☐ Mobile phone calls/messages
- ☐ Shouting/signaling
- ☐ Other (please specify)

10. What actions are taken when crowd congestion is observed? (Select all that apply)

- ☐ Directing people to less crowded areas
- ☐ Temporarily stopping entry
- ☐ Alerting higher authorities or supervisors
- ☐ Using physical barriers or signage
- ☐ Other (please specify)

11. How is crowd incident data or logs maintained?

- ☐ Written reports immediately after events
- ☐ End of shift summaries
- ☐ Verbal hand-offs without documentation
- ☐ No formal records were kept

12. How often are crowd management protocols reviewed or updated?

- ☐ Monthly
- ☐ Quarterly
- ☐ Annually
- ☐ Rarely or never

Open-ended questions for crowd monitoring personnel

1. What difficulties or challenges do you face when counting joggers manually at the CCSC oval?
2. Can you describe any specific challenges in managing jogger flow and crowd movements during peak times?
3. Are there any issues or limitations with the current manual people-counting procedures that should be addressed?
4. What improvements do you think could be made to make people counting more accurate or efficient?
5. How do you currently communicate and coordinate with other staff when monitoring the joggers?

6. What additional resources or support would help you better manage the crowd and joggers?
7. Are you willing to try or adopt new technologies (such as automated counters or monitoring devices) to assist in people counting and crowd management?
☐ Yes
☐ No
8. If willing, which technologies or tools would you prefer or find most helpful for crowd monitoring?

Jogger Questionnaire

Statistical Questions

1. How many times per week do you usually jog on the oval?
2. On which days of the week do you most often use the oval for jogging?
3. What time of day do you typically jog on the oval?
4. How long (in minutes or hours) do you usually spend jogging per session?
5. Which lanes do you most frequently use (e.g., inner, middle, or outer lanes)?
6. On average, how crowded do you perceive the oval to be during your usual jogging time?
7. Have you ever observed staff or devices counting or monitoring the number of joggers on the oval? (Yes/No)
8. If yes, at what times or locations do you usually notice these counting activities?

Qualitative Questions

Please rate your level of satisfaction with each aspect of your jogging experience on the CCSC oval by selecting the number that best reflects your feelings.

Satisfaction Scale:

1 – Very Dissatisfied

2 – Dissatisfied

3 – Neutral

4 – Satisfied

5 – Very Satisfied

1. I find the crowd density on the oval appropriate during my usual jogging times.
2. There is enough space on the oval for me to jog comfortably without feeling crowded.
3. Other joggers generally follow proper track etiquette, such as lane discipline and passing rules.
4. The rules and guidelines for using the oval are clearly communicated to me.
5. The overall jogging environment on the CCSC oval meets my expectations.
6. The condition of the oval supports my ability to maintain consistent jogging performance.¹
7. I am aware that when people counters (such as cameras or sensors) are used to monitor jogger occupancy.

8. The people counters used to monitor crowding help improve the management of the oval.
9. Receiving real-time updates about crowding or congestion on the oval would help me plan my jogging.
10. I would use alerts or notifications about peak crowd times if they were available.

CURRICULUM VITAE



ERIC MAR A. GEONZON

Minglanilla, Cebu

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Personal Information

Age: 20 years old

Gender: Male

Birthdate: July 9, 2004

Birthplace: Minglanilla, Cebu

Status: Single

EDUCATIONAL BACKGROUND:

Tertiary:

University of San Jose - Recoletos

2022 - Present

Secondary:

Don Bosco Technical College - Cebu Inc.

2016 - 2022

Elementary:

Sae Young Christian School

(2013). - 2016

Pardo Extension Elementary School

2007 - 2013



CHASHMYR IGNACIO

Bulacao Pardo, Cebu City

chashmyr.ignacio.22@usjr.edu.ph

Personal Information

Age: 21 years old

Gender: Female

Birthdate: February 04, 2004

Birthplace: Roxas City

Status: Single

EDUCATIONAL BACKGROUND:

Tertiary:

University of San Jose - Recoletos

2022- Present

Secondary:

Southwestern University PHINMA

2020 - 2022

Abellana National School

2016 - 2020

Elementary:

Cebu City Central School

2015 - 2016

Our Lady of Mt. Carmel Learning Center

2009 - 2015

Roxas International Pre-school

2007 - 2009



MYKYLLA ANGELYKA O. JORDAN

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Personal Information:

Age: 22 years old

Gender: Female

Birthdate: February 6, 2003

Birthplace: USA

Status: single

EDUCATIONAL BACKGROUND:

Tertiary:

University of San Jose-Recoletos	2022 - present
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Secondary:

Asian College of Technology	2019 - 2021
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Montessori Academy of Southern Cebu Inc.	2016 - 2019
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BD Child Care and Learning Center	2015 - 2016
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Elementary:

BD Child Care and Learning Center	2013 - 2015
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Leaton Academy	2012 - 2013
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St. Cecilia's College	2010 - 2012
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St. Paul College Foundation, Inc.	2009 - 2010
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